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Triply periodic minimal surfaces as catalyst carriers of compact reformers for hydrogen production and storage systems: recent advances and future perspectives

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HIGHLIGHTS

- TPMS-based catalyst carriers as a pathway to compact reformers.
- TPMS reformers boost thermal efficiency by 9.5–35.4%.
- TPMS structures provide superior heat-mass transfer vs. traditional designs.
- Additive manufacturing enables production of complex TPMS structures.

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ABSTRACT

The pursuit of compact, efficient, and high-performance reformers is crucial for advancing hydrogen production and storage technologies, particularly for on-board applications and thermochemical exhaust heat recuperation. Triply Periodic Minimal Surfaces (TPMS) have emerged as a novel class of lattice structures offering exceptional properties for process intensification. This comprehensive review systematically examines recent advances and future potential of TPMS-based structures as compact reformers for endothermic processes including steam methane reforming, methanol steam reforming, and ammonia decomposition.

Analysis reveals that TPMS-based reformers achieve thermal efficiency improvements of 9.5%–35.4% over conventional designs. Key findings include: (i) ammonia conversion of 75% at 450 °C using Ru-catalyzed Gyroid structures; (ii) Nusselt numbers 63%–96% higher than straight-tube configurations; (iii) pressure drop reductions of 50%–90% compared to packed beds; and (iv) hydrogen production rates of 5000 mL/min in optimized microreactors.

Despite this potential, industrial implementation faces significant hurdles: (a) manufacturing complexities in additive manufacturing, powder removal, and quality assurance; (b) coating challenges due to thermal expansion mismatches between metallic substrates ($15–20 \times 10^{-6}\,K^{-1}$) and ceramic catalysts ($5–10 \times 10^{-6}\,K^{-1}$); (c) computational limitations in multi-physics simulation; and (d) fabrication costs 10–100 times higher than conventional methods.

Future research priorities include: developing scalable coating methods for complex geometries; establishing long-term durability under realistic operating conditions; creating reduced-order models for rapid design optimization; and demonstrating TPMS reformers in target applications including on-board hydrogen production and exhaust heat recovery systems. Addressing these challenges will position TPMS structures as transformative platform technology for next-generation compact chemical reformers essential to the hydrogen economy.

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Nomenclature

Abbreviations

CFD	Computational fluid dynamics
$DMLS$	Direct metal laser sintering
$GHSV$	Gas hourly space velocity
HTC	Heat transfer coefficient (W m^{-1} K^{-1})
$LHHW$	Langmuir–Hinshelwood–Hougen–Watson technique
MSR	Methanol steam reforming
Nu	Nusselt number
$OHTC$	Overall heat transfer coefficient (W m^{-1} K^{-1})
Pr	Prandtl number
Re	Reynolds number
SLA	Stereo-lithography
SLS	Selective laser sintering
$TPMS$	Triply periodic minimal surface
TKE	Turbulence kinetic energy

Greek Symbols

α	Fusion coefficient
β	Geometric parameter in the kinetic term
β_w	Wrinkle factor
ϵ	Voidage
μ	Dynamic viscosity of the fluid (Pa s)
ϕ	Sphericity
ρ	Fluid density (kg m^{-3})
τ	Tortuosity
ε	Porosity

Roman Symbols

ΔD_p	Particle diameter deviation
ΔP	Pressure drop (Pa)
a	Edge length of the unit cell
D_p	Particle diameter (m)
f	Fanning friction factor
H	Mean curvature
k_1	Principal curvature
k_2	Principal curvature
L	Length of the porous medium (m)
n	Empirical coefficient
V	Superficial velocity (m s^{-1})

1. Introduction

Recent advancements in additive manufacturing, particularly in three-dimensional (3D) printing, have facilitated the fabrication of complex structures for various engineering applications [1–3]. Moreover, advanced computational design tools have been developed to enable the creation of geometrically complicated configurations, such as lattice structures – a design approach using repeated unit cell arrays. Consequently, these developments have significantly reduced fabrication limitations and enabled the implementation of lattice structures in various fields of engineering [4,5].

Based on these advancements, lattice structures have been increasingly utilized in various industries, including aerospace, biomedical, robotics, chemical, and thermal engineering [6–8]. These structures can be classified into two main subgroups – strut-based and triply periodic minimal surface (TPMS)-based configurations – depending on the unit cell design. These design differences enable modification of the mechanical, thermal and fluid properties of such structures to meet specific demands [9,10]. The adjustment of these parameters makes them suitable for applications ranging from lightweight load-bearing components to energy-efficient heat exchangers.

Among these, the strut-based cells, commonly referred to as lattice-frame materials, are composed of interconnected struts arranged in repeating unit cells [11,12]. The mechanical and thermal properties of these structures are determined by the geometric parameters of the unit cells, including their size, shape, and arrangement [13,14]. Typically, researchers have focused on the influence of unit cell geometry, with computational and experimental methods applied to analyze performance characteristics [15].

TPMS structures are a class of lattice structures defined by infinite, non-self-intersecting surfaces that are periodic along three orthogonal directions. Mathematically, TPMS are minimal surfaces, meaning that their mean curvature (H) is zero at every point:

$$H = \frac{k_1 + k_2}{2} = 0, \tag{1}$$

where k_1 and k_2 are the principal curvatures, representing the maximum and minimum normal curvatures, respectively.

The curvature characteristics of the TPMS reveal their fundamental geometric properties. As visualized in Fig. 1, the mean curvature H vanishes identically ($H \equiv 0$) across all structures. Although the uniform color representation in Fig. 1 may seem monotonous, it shows one of the main geometric properties of TPMS. In particular, condition $H = 0$ indicates that at any given point the principal curvatures must be equal in magnitude but opposite in sign ($k_1 = -k_2$). Fig. 1 clearly illustrates this balance: every point on these surfaces forms an ideal shape. This ensures a smooth transition of Primitive, Gyroid, and I-WP geometries, among others, in three-dimensional space without sharp edges, vertices, or abrupt geometric discontinuities. At the same time, Fig. 2 presents the corresponding Gaussian curvature distributions, which exhibit non-zero but periodic variations across each surface type. The contrast between the constant mean curvature (Fig. 1) and the spatially varying Gaussian curvature (Fig. 2) highlights the distinctive differential geometry of TPMS. These geometric properties – combined with their inherent periodicity and infinite extendibility – make TPMS particularly valuable for engineering applications that require optimized surface-to-volume ratios or customized mechanical performance [16–18].

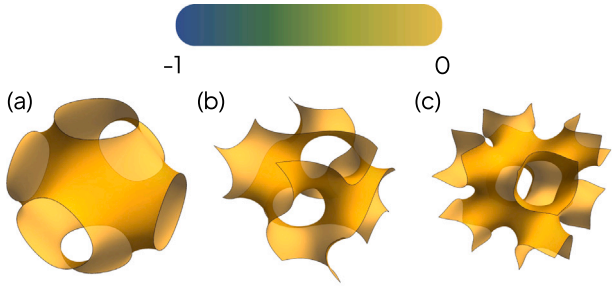


Fig. 1. Mean curvature distribution for TPMS structures: (a) Primitive, (b) Gyroid, and (c) I-WP surfaces.

Building on these geometric properties, TPMS structures exhibit superior performance characteristics compared to traditional designs such as plate, tubular, and honeycomb structures. For example, these surfaces eliminate stress concentrations through their smooth curvature, enhancing stiffness and enabling a uniform load distribution [19–21]. Furthermore, their high surface-to-volume ratio and ordered, non-tortuous pore networks significantly improve heat transfer, mass transport, and thermal conductivity [22,23]. The optimized flow paths in TPMS structures reduce pressure losses while maintaining efficient fluid transport, making them particularly suitable for heat exchangers and chemical reactors [24,25]. The compatibility of TPMS structures with

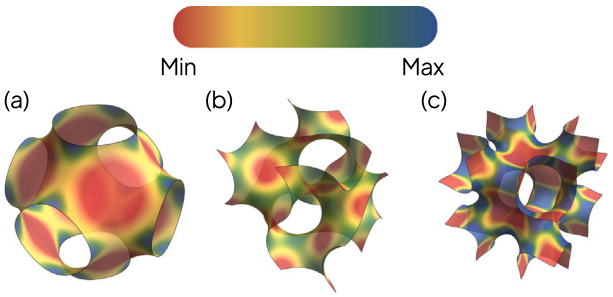


Fig. 2. Gaussian curvature distribution for TPMS structures: (a) Primitive, (b) Gyroid, and (c) I-WP surfaces.

additive manufacturing techniques facilitates the production of these complex geometries for practical engineering applications [26].

TPMS-based structures demonstrate high efficiency in thermal engineering applications. Torri et al. reported improvements in the heat transfer rate of up to 55% compared to conventional designs, attributed to their expansive surface areas and low pressure drops [27]. The inherent compactness of TPMS structures enables size reductions of 3–10 times relative to traditional tubular heat exchangers, while maintaining superior cooling performance in high-heat environments.

The thermal advantages of TPMS structures extend to radiator designs for electronic devices, where efficient heat dissipation is critical. These structures enhance convective heat transfer by optimizing surface area exposure to cooling fluids. Yinzheng et al. demonstrated that TPMS-based radiators outperform traditional designs by up to 27% under turbulent flow conditions [28], a particularly valuable improvement for space-constrained compact electronics.

TPMS structures contribute significantly to advances in membrane distillation and hydrogen storage technologies. Kumar et al. reported a 60% improvement in heat and mass transfer efficiency for membrane distillation applications compared to conventional designs [29]. In hydrogen storage systems, TPMS structures enhance sorption kinetics, enabling more efficient energy storage solutions as documented by Lesmana et al. [30]. These applications demonstrate how TPMS-based designs combine both thermal and fluid dynamic benefits.

Additionally, TPMS-based structures can be used in chemical reactor applications, particularly for catalytic processes. In the investigation of hydrogen production via ammonia decomposition, Italiano et al. [31] revealed that gyroidal TPMS structures achieve significantly higher conversion efficiency at moderate temperatures (400–500 °C) compared to conventional packed bed reactors. The authors demonstrated three key advantages of the TPMS configurations: enhanced mass transfer through optimized fluid pathways, substantial reduction in pressure drop, and high decomposition efficiency reaching 99% at 600 °C. These combined characteristics establish TPMS-based reactors as a compact and energy-efficient solution for ammonia decomposition, offering improved reaction kinetics while maintaining structural integrity under operational conditions. The geometric optimization inherent in TPMS designs enables more efficient catalyst utilization and better thermal management compared to traditional reactor geometries.

The aforementioned examples demonstrate the superior thermal and fluid dynamic performance of TPMS-based structures, which are crucial for various chemical reformers in endothermic processes. These structures show significant potential for the development of efficient and compact chemical reformers that could be integrated into energy conversion and storage systems, as well as chemical and environmental engineering applications. Despite this promising direction, research in this field remains limited.

The primary objective of this review is to provide a comprehensive overview of the application potential of TPMS structures as advanced catalyst carriers in compact reformers, specifically for hydrogen production and storage systems. The performance of such reformers is governed by three interdependent factors: fluid flow characteristics, heat transfer rates, and mass transfer effectiveness. This work systematically organizes and analyzes existing research on TPMS applications in compact chemical reformers for endothermic processes. In particular, it focuses on the interrelationships between flow dynamics, thermal transport phenomena, and chemical reaction kinetics.

To ensure a comprehensive and objective analysis of the current state of research on TPMS-based catalyst carriers and heat exchangers, a systematic literature search was conducted. The primary literature search was performed using major scientific databases, including Scopus and Web of Science. Generally, the search was restricted to papers published between 2024 and 2026 to capture the most recent advances in TPMS fields. To maintain the high quality and specific focus of this review, strict inclusion and exclusion criteria were applied. Suitable publications included peer-reviewed scientific articles and comprehensive reviews published in English that specifically investigated the thermal, hydraulic, or catalytic characteristics of TPMS structures. In addition, selected articles had to contain quantitative experimental or numerical data on parameters such as heat transfer coefficients, pressure drop, Nusselt numbers, or reaction rates. Conversely, studies devoted exclusively to biomedical, orthopedic, or acoustic applications, such as bone scaffolds, were excluded, as were mathematical or topological studies, as they fall outside the scope of chemical engineering and energy conversion.

This is the first review dedicated to the application of TPMS-based structures specifically as catalyst carriers as shown in Table 1. By synthesizing and systematizing the majority of contemporary scientific papers on this topic, the presented review specifically analyzes how TPMS design parameters influence the performance of both heat exchangers and chemical reformers. Through a comparative evaluation with conventional designs, the distinct advantages of TPMS structures in improving heat-mass transfer efficiency and fluid dynamics performance were highlighted. This work provides valuable information on recent technological advances. Furthermore, it identifies promising research directions for implementing TPMS structures in next-generation compact reformers used in energy storage and conversion systems.

2. TPMS fundamentals

This section provides a geometric characterization of TPMS, exploring their fundamental mathematical properties. It includes a historical retrospective that traces the evolution of these structures from abstract theoretical concepts to practical engineering applications.

Table 1
Comparative analysis of Physics Fields in TPMS literature.

Physics Fields / Applications	Current review	Ifa and Efa [32]	Wang et al. [33]	Shingare et al. [34]	Samudin et al. [35]
Geometry analysis	+	+	+	+	+
Flow Dynamics	+	–	+	–	–
Heat Transfer	+	+	+	–	–
Mass Transfer	+	–	+	–	–
Catalysis / Reaction Kinetics	+	–	–	–	–
Mechanical Properties	–	–	–	+	+

Note: '+' indicates the field is significantly covered; '-' indicates it is absent or negligible.

2.1. Historical development of TPMS structures

The development of TPMS began in the 19th century with the mathematical works of Hermann Schwarz [36]. Schwarz introduced three minimal surfaces – Schwarz P, Schwarz D, and Schwarz H – and was the first to characterize them as surfaces with zero mean curvature. Together with his student Edvard Neovius, Schwarz expanded the theoretical framework of TPMS, focusing on their mathematical properties [37]. However, the works of Schwarz and Neovius remained purely theoretical, as industrial applications were impossible at the time due to the lack of manufacturing technology capable of producing such complex structures.

In the 1970s, research on TPMS advanced significantly through the contributions of Alan Schoen [38], who developed novel TPMS structures and demonstrated their applicability in modeling natural systems such as crystals, soap foams, and biological membranes. Subsequent research in the 1980s and 1990s further explored the properties of TPMS; yet, practical implementation remained unfeasible due to persistent manufacturing limitations [39]. These foundational studies paved the way for modern engineering applications of TPMS.

Significant progress in TPMS research and applications was achieved in the 2000s, driven by advances in additive manufacturing techniques such as 3D printing and laser powder cladding, which enabled the fabrication of complex TPMS-based structures with high precision. By the 2010s, research had expanded into materials science and biomedicine [40,41], with TPMS widely used to design lightweight, high-strength materials for prostheses, implants, and other applications. This demonstrates the transition of TPMS from theoretical constructs to practical engineering solutions.

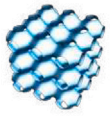
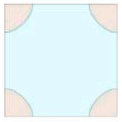
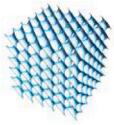
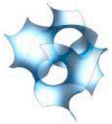
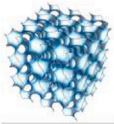
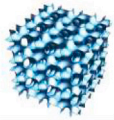
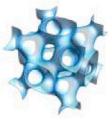
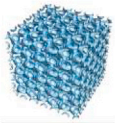
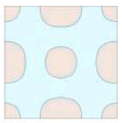
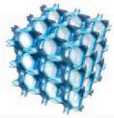
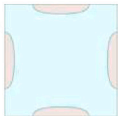
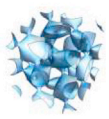
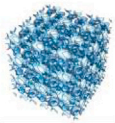
The historical overview above illustrates the evolution of TPMS from abstract mathematical concepts to functional engineering systems. To clarify the scope and diversity of TPMS structures—including the fundamental Schwarz P, D, and H surfaces, among others—a systematic classification is necessary. Table 2 summarizes the main TPMS types, providing their nomenclature, mathematical formulas, surface-to-volume ratios, and visual representations. Note that exact surface-to-volume ratios are listed only for Schwarz P, Schwarz D, Gyroid, and I-WP, as these are the only surfaces for which analytical solutions exist. For other structures, such as F-RD, Neovius, and Lidinoid, the ratios were computed numerically using Wolfram Mathematica.

2.2. TPMS design advantages

Based on the geometric and topological characteristics, TPMS provide significant performance advantages over conventional honeycomb, tubular, and plate-based designs. These structures combine a high surface-to-volume ratio, periodic topology with minimal curvature, and a strong lattice structure, making them suitable for lightweight applications requiring high thermal efficiency and fluid permeability [42–44].

The most significant property of TPMS is its high surface-to-volume ratio, which promotes efficient heat and mass transfer by maximizing

Table 2
Mathematical and geometric characterization of some triply periodic minimal surfaces.

TPMS	Mathematical formula	S-to-V ratio	Unit cell	Full structure	Slice
Schwarz P (Primitive)	$\cos(x)+\cos(y)+\cos(z)=0$	$\frac{2.345}{a}$ [38]			
Schwarz D (Diamond)	$\sin(x)\sin(y)\sin(z)+$ $+\sin(x)\cos(y)\cos(z)+$ $+\cos(x)\sin(y)\cos(z)+$ $+\cos(x)\cos(y)\sin(z)=0$	$\frac{2.418}{a}$ [38]			
Gyroid	$\sin(x)\cos(y)+\sin(y)\cos(z)+$ $+\sin(z)\cos(x)=0$	$\frac{2.453}{a}$ [38]			
I-WP	$2[\cos(x)\cos(y)+\cos(y)\cos(z)+$ $+\cos(z)\cos(x)]-$ $-[\cos(2x)+\cos(2y)+\cos(2z)]=0$	$\frac{3.260}{a}$ [39]			
F-RD	$\cos(2x)+\cos(2y)+\cos(2z)-$ $-\cos(x)\cos(y)-\cos(y)\cos(z)-$ $-\cos(z)\cos(x)=0$	$\frac{5.068}{a}$			
Neovius	$\cos(x)+\cos(y)+\cos(z)+$ $+4\cos(x)\cos(y)\cos(z)=0$	$\frac{4.113}{a}$			
Lidinoid	$\sin(2x)\cos(y)\sin(z)+\sin(2y)\cos(z)\sin(x)+$ $+\sin(2z)\cos(x)\sin(y)-\cos(2x)\cos(2y)-$ $-\cos(2y)\cos(2z)-\cos(2z)\cos(2x)+0.5=0$	$\frac{4.332}{a}$			

Note: a - edge length of the unit cell.

the contact area between the solid and liquid phases [45]. In literature, this ratio is typically quantified in two ways: relative to the enclosed volume of the structure itself [46,47], or relative to the volume of the bounding unit cell [48,49].

Figs. 3 and 4 illustrate these metrics for a Schwarz P structure. In the first case (Fig. 3), the surface-to-volume ratio is shown as a function of wall thickness in various cell sizes. Although a traditional flat plate may exhibit a higher ratio at very small thicknesses (< 0.78 mm), the Schwarz P structure consistently outperforms it as the thickness increases (> 0.9 mm). In the second case (Fig. 4), where the ratio is normalized by the bounding cube volume, the flat plate remains constant, whereas the TPMS structures demonstrate a rapid increase in surface area as the unit cell size decreases. These findings confirm that TPMS structures are not only competitive, but can significantly exceed the performance of planar structures.

Beyond surface area, the smooth, continuous curvature, and complex internal channels of TPMS make intense fluid mixing and generate turbulence. This mechanism disrupts the thermal and hydrodynamic boundary layers, leading to higher transport coefficients. For instance, Diamond and Gyroid structures promote internal fluid rotation and boundary layer displacement [50,51], enhancing turbulent mixing in applications with high heat-flux demands [52,53]. Additionally, the continuous flow channels minimize stagnation zones and optimize pressure drop characteristics compared to traditional geometries with sharp edges [54,55].

From a structural perspective, TPMS achieve a comprehensive three-dimensional structure using minimal material, which substantially reduces overall weight [56,57]. The minimal surface geometry ensures a uniform distribution of mechanical loads, preventing localized stress concentrations typically found in discrete lattice or plate designs. This results in a combination of high specific strength and lightweight properties [58,59]. In addition, TPMS parameters such as wall thickness and cell size can be adapted to specific local conditions. For example, by strengthening areas subject to high thermal or mechanical stresses while maintaining a low density in other parts to optimize weight efficiency [47,60].

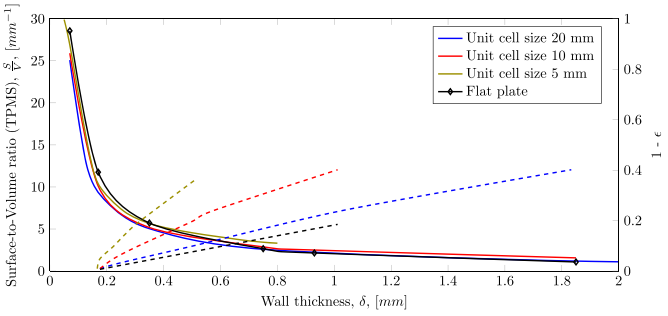


Fig. 3. Surface-to-volume (TPMS) ratio and voidage as a function of wall thickness for Schwarz P structure: line – surface-to-volume, dashed lines – voidage.

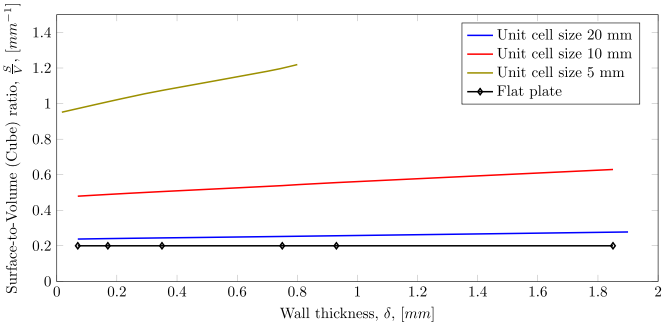


Fig. 4. Surface-to-volume (Cube) ratio as a function of wall thickness for Schwarz P structure.

3. Manufacturing

This section describes the role of additive manufacturing in TPMS application. In addition, it outlines research tasks for the potential application of TPMS-based structures in compact chemical reformers for endothermic processes.

3.1. Additive manufacturing and TPMS

Additive manufacturing, or 3D printing, has emerged as a prominent technology for fabricating complex TPMS-based structures. This technique enables the layer-by-layer fabrication of geometrically intricate shapes with high precision. The advancement of additive manufacturing has facilitated the use of various materials and the creation of TPMS architectures, which were previously unattainable using conventional manufacturing methods [61,62].

Traditional manufacturing techniques, such as casting or machining, are unsuitable for producing TPMS structures due to several inherent limitations. These include the inability to accurately replicate complex periodic geometries, the high costs associated with small-batch production, and significant material waste. Such constraints render traditional approaches economically and technically impractical. In contrast, additive manufacturing overcomes these limitations by enabling layer-by-layer fabrication of complex geometries with minimal restrictions.

Additive manufacturing offers numerous advantages for the production of TPMS structures. Techniques such as stereo-lithography (SLA) (Fig. 5), selective laser sintering (SLS), and direct metal laser sintering (DMLS) provide high precision and compatibility with a wide range of materials, from polymers to metals (Table 3). Furthermore, 3D printing allows for the creation of structures with controlled porosity and functionally graded properties tailored to specific applications. Additionally, additive manufacturing significantly reduces material waste, which is particularly advantageous for high-value alloys. These benefits make



Fig. 5. 3D printing of a TPMS-based structure via the SLA method.

Table 3
Material classification for 3D-printed TPMS structures.

Material	Printing methods	Properties	Temperature limit	Applications
Polymers	FDM, SLA, DLP	Lightweight, Low cost, Limited strength	60–100 °C	Flow dynamics, [72] Heat transfer [73]
Composites	FDM, SLA	Lightweight, High strength	>150 °C	Flow dynamics, [73] Heat transfer [73]
Metals	SLM, DMLS, EBM	High strength, Corrosion resistance	>1000 °C	Flow dynamics [74,75] Heat transfer [74,75] Chemical reactions [31]
Ceramics	SLA with suspensions, Binder Jetting, Sintering	Brittle, Chemical resistance	>1600 °C	Catalyst supports [76], Heat transfer [77], Mechanical properties [78]

3D printing an ideal method for producing TPMS structures, thereby expanding their applicability across various industries.

Despite these possibilities, the manufacture of TPMS structures is associated with significant practical difficulties [63]. Complex, continuous internal voids make post-processing very problematic. For example, in powder layer melting processes, powder jamming in cellular channels makes thorough cleaning extremely difficult [64]. In addition, the internal support structures often required for complex protrusions are difficult to design and virtually impossible to remove from the finished part. Finally, verifying the structural integrity of the manufactured TPMS is a major obstacle. Since internal struts and walls cannot be inspected using conventional visual inspection methods, quality assurance relies heavily on costly and labor-intensive micro-computed tomography (micro-CT) scans to detect internal voidage or manufacturing defects [65,66].

Consequently, TPMS structures manufactured using additive manufacturing are increasingly employed in aerospace, biomedical, and energy applications. In aerospace, they are utilized to manufacture lightweight yet robust components, such as wing elements and turbine blades, from titanium alloys, aluminum alloys, and stainless steel, thus reducing weight and fuel consumption [67,68]. In biomedicine, porous TPMS structures composed of bio-compatible materials such as titanium, stainless steel, polycaprolactone, polyetheretherketone, and beta-tricalcium phosphate are used to produce implants that promote osseointegration due to their controlled porosity [69,70]. In the energy sector, TPMS-based heat exchangers made of aluminum and stainless steel enhance thermal effectiveness owing to their high surface-to-volume ratio. Additive manufacturing enables the optimization of these structures for specific functional requirements that balance mechanical strength, weight, and other critical properties.

Moreover, the integration of modern engineering software has streamlined additive manufacturing processes. Currently, numerous software tools, such as SolidWorks, Ansys, and Blender, are available to design and analyze surface and solid geometries [71]. These tools expedite the development and evaluation of TPMS structures while enabling comprehensive investigations, from geometric modeling to finite element method (FEM) analysis. Such advances have significantly accelerated the design and analysis of TPMS architectures.

However, in practice, creating and investigating TPMS structures involves serious computational difficulties. Standard CAD environments are optimized for representing boundaries rather than implicit mathematical formulas. Consequently, TPMS modeling requires the conversion of continuous equations into discrete polygonal meshes. Usually, such tessellation creates serious problems with point data management, resulting in large file sizes. In addition, these high density meshes very often generate artifacts, which make the transition to FEM analysis extremely difficult due to mesh generation failures and long computation times.

Thus, additive manufacturing has become a pivotal technology for TPMS structure fabrication. However, the high cost of equipment remains a substantial barrier, limiting large-scale industrial adoption. Additionally, current printing speeds are insufficient for mass production, restricting scalability. Despite these challenges, the future of additive manufacturing in TPMS applications appears highly promising, driven by continuous advances in material science and process optimization. Further developments, coupled with emerging technologies such as artificial intelligence (AI)-assisted design, are expected to enhance precision and efficiency, paving the way for broader utilization of TPMS structures.

3.2. Prospects of TPMS in catalytic reformers

Catalytic chemical reformers are widely used in chemical engineering, energy storage and energy conversion systems, particularly for steam methane reforming, methanol synthesis, and the ammonia decomposition process [79–81]. In such reformers, mass transfer plays a crucial role alongside heat transfer efficiency and flow dynamics. Traditional solutions include packed beds of catalyst particles [82], honeycomb structures [83,84], plate-coated reformers [85,86], and catalytically activated foam structures [87,88]. The surface-to-volume ratio is a key parameter determining reformer performance and size.

The importance of a high surface-to-volume ratio stems from its impact on heat and mass transfer within the catalyst particle or layer. Intra-particle diffusion significantly limits the efficiency of catalytic reformers [89]. Pedernera et al. reported that only a thin layer (approximately 3% of the catalyst) contributes substantially to overall conversion in catalytic reformers [90]. Furthermore, previous studies have demonstrated that higher surface-to-volume ratios in catalytic reformers lead to improved conversion rates of initial components [91].

To overcome these limitations, TPMS structures have been proposed as innovative catalytic supports for chemical reformers. By applying a uniform catalytic coating to the TPMS framework, diffusion limitations are minimized, and the entire surface area becomes available for chemical reactions. This approach optimizes catalyst utilization by enabling reactions across the full surface area, in contrast with the surface-limited activity observed in packed beds. Additionally, the inherent surface-to-volume ratio characteristics of TPMS facilitate the development of compact reformer designs. This size reduction improves space efficiency while simultaneously decreasing material requirements and operational costs, making TPMS particularly suitable for space-constrained applications.

The defining characteristic of TPMS structures is their ability to divide a unit cell volume into two non-intersecting channels. This feature makes them particularly advantageous for reformers where heat for endothermic processes is supplied through the reformer wall from hot steam. TPMS offers a unique solution for compact reformers requiring

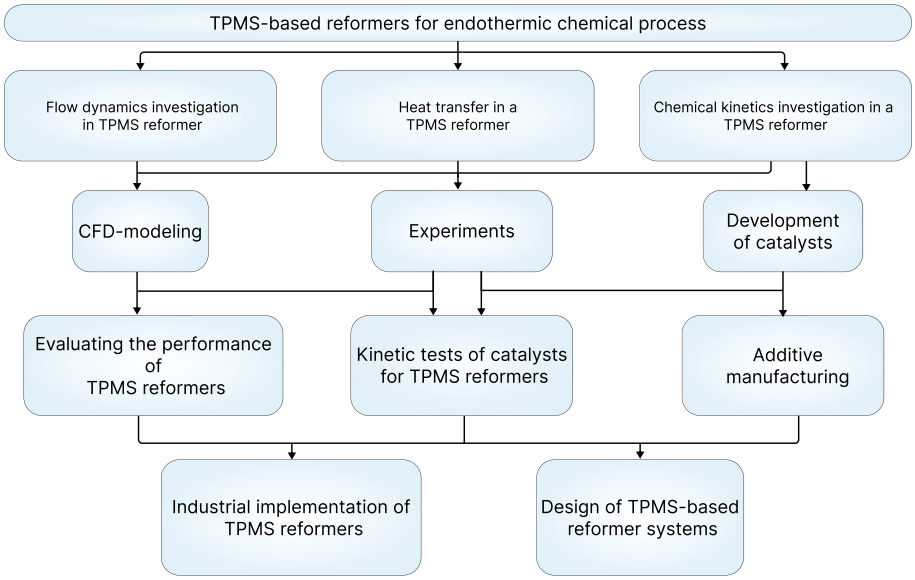


Fig. 6. Research framework for investigating TPMS-based reformers in endothermic chemical processes.

heat transfer from an adjacent stream, such as in thermochemical exhaust heat recuperation systems [92–94].

For compact catalytic reformers, TPMS structures offer several key advantages: enhanced overall heat transfer coefficients, superior surface-to-volume ratios, and reduced pressure drops. Considering current advancements in additive manufacturing, the implementation of TPMS in compact reformers presents not only a scientifically promising approach but also a significant potential for industrial application. Fig. 6 presents a schematic representation of research tasks related to the investigation and industrial implementation of TPMS in compact chemical reformers.

Current research findings underscore the potential of TPMS structures to enhance chemical reaction efficiency. Their high surface-to-volume ratio, uniform flow distribution, and excellent heat transfer characteristics make them effective as catalyst supports and heat exchangers in compact chemical reformers. To fully understand the impact of TPMS structures on process efficiency, further investigation of their flow dynamics is essential, as this represents a critical factor in determining their performance in chemical engineering applications.

4. Flow dynamics

Traditionally, researchers have investigated flow dynamics through computational fluid dynamics (CFD), experimental tests, and analytical methods, either separately or in combination. However, studies on flow dynamics in TPMS-based structures are typically presented in the literature as integrated investigations [95,96]. Consequently, this section is not subdivided by methodological approaches (experimental vs. computational), but rather focuses on macroscopic influencing factors. The primary emphasis of this section is to examine the effects of key design and operational parameters – including TPMS topology, unit cell size, porosity, and flow velocity – on the pressure drop and friction factor characteristics of TPMS-based structures. For the first time, we systematically consolidate and analyze the relationship between Reynolds number (Re) and friction factor (f) for TPMS structures. Our findings reveal that the friction factor dependence on the Reynolds number in TPMS structures follows a trend analogous to the fundamental correlation established by Nikuradse [97].

4.1. Pressure drop

The design of compact chemical reformers necessitates an optimal balance between heat transfer performance, pressure drop characteristics, and physical dimensions. A detailed understanding of the trade-offs between heat transfer efficiency and hydraulic resistance requires systematic analysis of the pressure drop in various TPMS-based structures [98,99]. Recent investigations have extensively examined the pressure drop under laminar flow conditions for different TPMS architectures [100,101]. While numerous studies have concurrently analyzed both heat transfer and pressure drop phenomena, research focusing exclusively on flow dynamics and pressure drop remains comparatively limited [102–104].

First, Zhao et al. conducted a comprehensive numerical investigation of heat transfer and pressure drop correlations in nine distinct TPMS structures [105]. Their analysis of flow characteristics and permeability revealed significant dependencies on both geometric configuration and unit cell dimensions. The study identified the Gyroid, Diamond, F-K S, Split P, and G Prime structures as having superior hydraulic performance. These structures exhibit reduced pressure drops due to their uniform flow cross-sections and enhanced permeability. The research demonstrated that permeability increases proportionally with unit cell size, but decreases with additional structural layers. Hybrid G-G P structures, combining Gyroid and G Prime flow characteristics, were evaluated across fusion coefficients α, with optimal performance observed at $\alpha = 2$. Increasing α to values of 4 and 6 resulted in permeability reductions of 13.65% and 14.94%, accompanied by pressure drop increases of 28.52% and 30.99%, respectively. Furthermore, the Darcy number exhibited an asymptotic behavior with increasing layer count, initially decreasing before stabilizing.

Renon and Jeannigros developed specialized correlations for Diamond-structured heat exchanger channels, establishing a Fanning friction factor relationship for fully developed flow conditions [106]:

$$f = 1.85 \cdot Re^{-0.17}. \tag{2}$$

This correlation, derived from 40 computational configurations, demonstrated an average relative error of 3% with a maximum deviation of 9%. Comparative analysis revealed the Diamond structure's superior hydraulic performance, with Gyroid configurations exhibiting 12% to 17% higher friction factors across all tested conditions. These findings suggest that Diamond structures may be preferable for applications where pressure drop minimization is critical.

The significant influence of TPMS geometry on hydraulic performance necessitates detailed analysis of pressure drop characteristics. Bragin et al. conducted a rigorous multi-method investigation of three

TPMS configurations: Schwarz P, Schoen I-WP, and Neovius [72]. Their study, which incorporated numerical simulation, experimental validation, and analytical modeling, revealed substantial variations in the pressure drop across geometries. Schwarz P demonstrated optimal hydraulic performance due to its tubular flow pattern (Fig. 7a), while Neovius exhibited the highest pressure drops resulting from intensive flow mixing. Schoen I-WP showed intermediate characteristics (Fig. 7b). The research identified significant limitations in conventional pressure drop equations, with the Ergun and Burke-Plummer relationships overestimating by approximately 100%, and the Kozeny-Carman equation underestimating by more than 1000% compared to experimental data. To address these discrepancies, a modified Hagen-Poiseuille equation incorporating both viscous and kinetic energy loss components was proposed and represented as:

$$\frac{\Delta P}{L} = \frac{72\mu\tau(1-\varepsilon)^2 V}{D_p^2\phi^2\varepsilon^3} + \frac{3\tau(1-\varepsilon)\rho V^2 n\left(\frac{3}{2}+\frac{1}{\rho^4}-\frac{5}{2\beta^2}\right)}{4\varepsilon^3 D_p\phi}, \tag{3}$$

where ΔP – pressure drop; L – length of the porous medium; μ – dynamic viscosity of the fluid; τ – tortuosity; ε – porosity; V – superficial velocity; D_p – particle diameter; ϕ – sphericity; ρ – fluid density; n – empirical coefficient; β – geometric parameter in the kinetic term.

Using this modified equation, the pressure drop was calculated and compared with CFD predictions, as shown in Fig. 7(c). The discrepancy between the CFD and analytical results underscores the inherent difficulty in predicting the pressure drop for TPMS structures due to their complex flow dynamics. This work clearly demonstrates the complexity and critical importance of accurate pressure drop prediction in TPMS-based designs.

Ribeiro et al. investigated a novel design approach for gas-liquid reactor packing using TPMS structures, with a focus on pressure drop characteristics [107]. They developed and evaluated three novel TPMS prototypes with varying strut diameters using CFD simulations, complemented by experimental validation of the gas-solid pressure drop in a small-scale domain. The CFD results demonstrated that the pressure drop is significantly influenced by strut diameter, with larger diameters resulting in higher pressure drops. For instance, simulations indicated that Prototype 1 (cavity size = 4.3 mm) exhibited a pressure drop 16% higher than Prototype 3 (cavity size = 6.6 mm), while the experimental results showed a difference of 17% to 24%. The authors suggested that the increased pressure drop associated with larger strut diameters may be balanced by improved gas-liquid interaction performance. They also established that the flow characteristics in TPMS are related to unit cell size, which can significantly increase the pressure drop.

Thus, existing studies generally confirm the influence of the TPMS geometry on the pressure drop. A recurring finding is that voidage and unit cell size are also critical factors influencing flow behavior. Therefore, the following subsection analyzes the specific effects of these two parameters on flow characteristics, particularly pressure drop.

Hawken et al. characterized the pressure drop in 3D-printed Diamond TPMS structures [108]. Using experimental measurements and X-ray computed tomography, they established that pressure drop decreases with increasing voidage. The study showed that existing

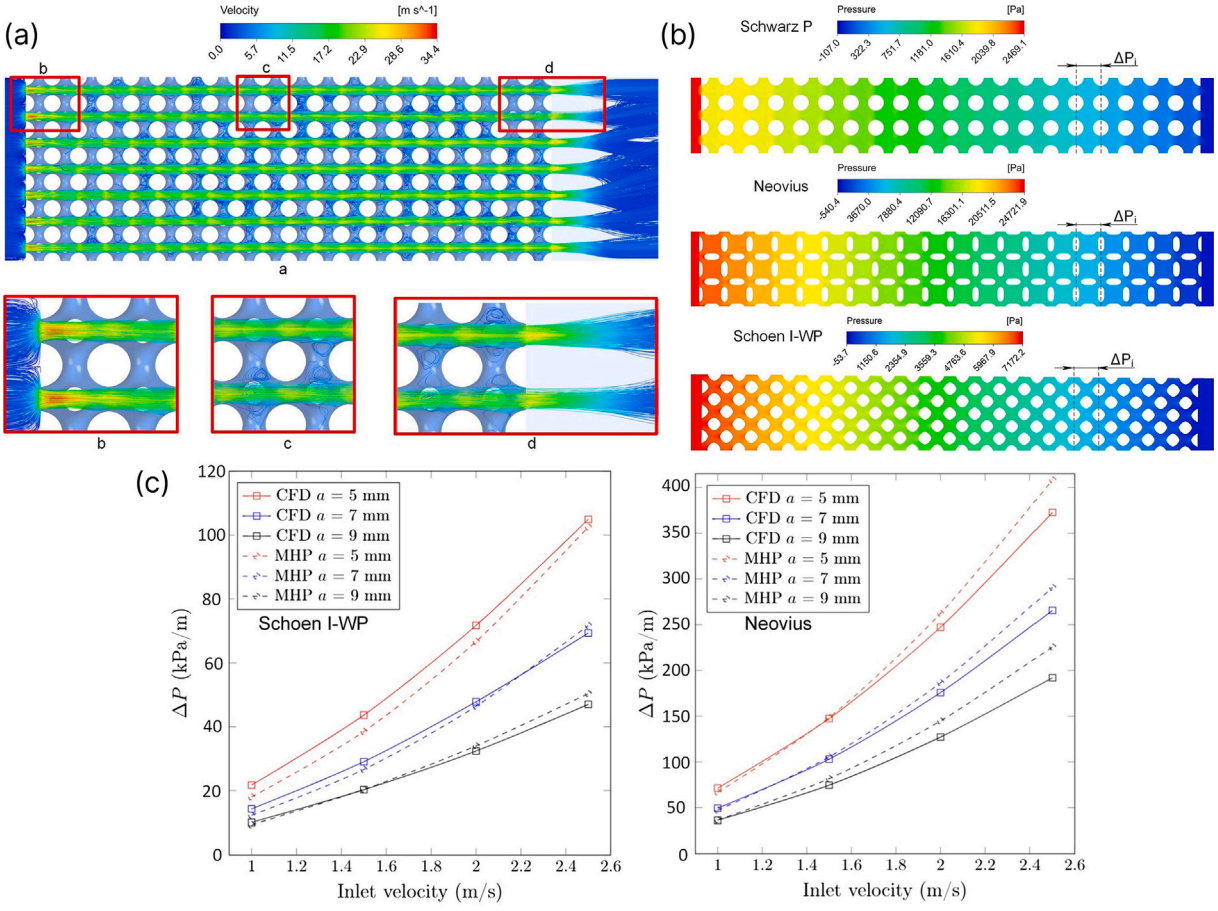


Fig. 7. Comparison of flow dynamics characteristics of different TPMS structures (according to Bragin et al. study): (a) side view of the flow distribution within a Schwarz P TPMS structure; (b) pressure drop along the packed bed length for all three structures; (c) comparison of pressure drop from CFD simulations and the modified Hagen-Poiseuille equation (Eq. 3) for Schoen I-WP and Neovius structures. All results are for an inlet velocity of 2 m/s, porosity of 0.35, and cell dimension of 5 mm [72].

correlations, such as the Ergun equation, did not accurately predict the pressure drop. Consequently, new correlation coefficients for the Ergun equation for the Diamond were proposed:

$$K_1^* = \frac{76.6}{\epsilon^{0.4}}; \quad K_2^* = \frac{0.91}{\epsilon^{0.5}}. \tag{4}$$

In practical application, the study determined that printer type and column diameter had no significant effect on pressure drop. Wall effects were also negligible, even at small aspect ratios, underscoring that the internal geometry is the dominant factor governing pressure drop behavior. These findings confirm that voidage significantly affects pressure drop in TPMS structures; however, the specific influence of unit cell size remained unquantified.

Sun et al. conducted a numerical analysis to evaluate the influence of voidage and unit cell size on the thermal-hydraulic performance of Gyroid TPMS heat exchangers [109]. Voidage was controlled by varying the unit cell size and wall thickness. The results demonstrated a substantial impact on pressure drop: at a voidage of 60%, the pressure drop was 413–694% higher on the hot side and 267–280% higher on the cold side compared to the structure with 80% voidage. Furthermore, the turbulence kinetic energy (TKE) was significantly higher at lower voidages, with the average TKE increasing by 255% when the voidage was reduced from 80% to 60%. These results conclusively show that voidage is a critical parameter governing the gas flow behavior and pressure drop in TPMS structures.

Shahid et al. computationally evaluated the influence of voidage on the flow properties of Diamond and Gyroid TPMS structures [110]. Their CFD results demonstrated that a higher voidage significantly reduces flow resistance and pressure drop, thus improving fluid transport, while a lower voidage increases mechanical strength at the cost of a higher pressure drop. Specifically, as the voidage increased from 50% to 80%, the pressure drop decreased from 2.079 to 0.984 Pa for the Diamond structure and from 1.669 to 0.943 Pa for the Gyroid structure. Although such studies clarify the role of voidage, the specific effects of unit cell size on flow behavior are poorly understood.

Zhang et al. directly investigated this by quantitatively analyzing the thermal convection performance in Gyroid structures with varying unit cell sizes at a constant voidage [111]. Using combined experimental and CFD methods, they found that reducing unit cell size increases main flow velocity and increases pressure drop by 26.3% to 36.4%. These results confirm that even at constant voidage, unit cell size significantly affects hydraulic performance.

In a related but distinct study, Xin Yu et al. experimentally evaluated the influence of unit cell size on the manufacturability and mechanical properties of TPMS structures fabricated by selective laser melting [112]. They reported that increasing the unit cell size reduces the deviation between the actual and designed voidage (from 9.15% to 3.10%) and increases the tensile strength up to a threshold, but decreases the load-bearing capacity. Conversely, smaller unit cell sizes improve load-bearing capacity at the potential cost of void accuracy. While this work provides valuable insights into mechanical properties, their relationship with other performance metrics requires further exploration.

The studies described above, involving various TPMS structures such as Gyroid, Diamond, Split P, Schwarz P, I-WP, and Neovius, consistently demonstrate that geometry is a critical determinant of pressure drop characteristics under laminar flow conditions. Fig. 8 systematizes experimental and numerical results on the pressure drop across these diverse TPMS-based structures.

As illustrated in Fig. 8, the relationship between the Reynolds number and the friction factor exhibits a similar decaying trend for various configurations of Diamond and Gyroid-based structures, characteristic of laminar and transitional flows. In contrast, Schwarz P structures consistently exhibit a lower friction factor, as shown in Fig. 8, indicating a superior hydraulic performance that makes them preferred

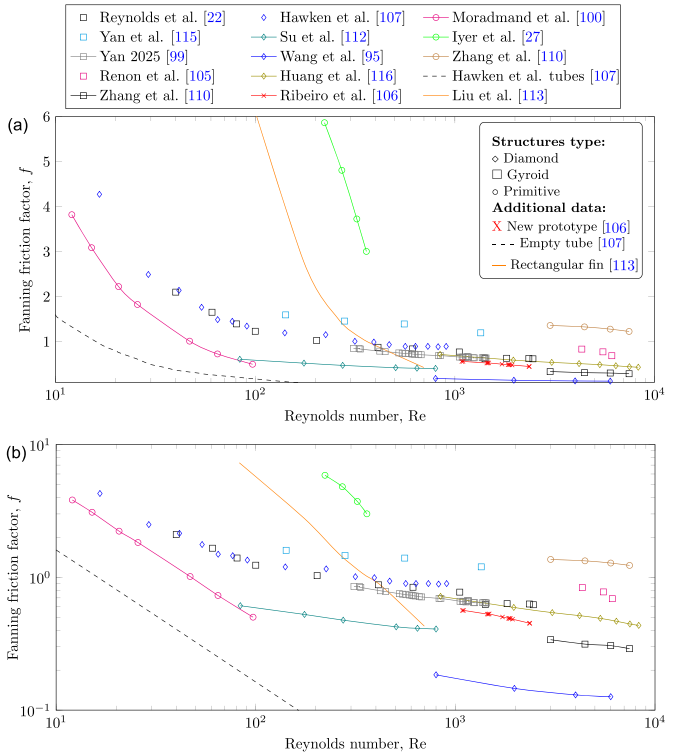


Fig. 8. Friction factor as a function of *Re*: line – numerical simulations, points – measured data. (a) Linear y-axis, (b) logarithmic y-axis.

for applications where minimizing energy loss is critical, such as in low-power fluidic systems.

Beyond the core topology, specific geometric parameters play a crucial role. Wall thickness and rod diameter significantly influence flow resistance; thicker walls and larger struts reduce the flow area and increase tortuosity, thereby elevating pressure drop. While this may be detrimental to hydraulic efficiency, it can be beneficial in enhancing distribution uniformity or intensifying gas-liquid interactions in reactor designs.

The geometric complexity of TPMS renders traditional porous media correlations, such as the Ergun equation, insufficiently accurate, often yielding errors exceeding 100%. This discrepancy has led to the development of new TPMS-specific analytical models and correlation coefficients to better predict hydraulic performance.

To date, research has been predominantly focused on laminar and low *Re* flows. However, studying the behavior of TPMS structures at high *Re*, where inertial effects and turbulence dominate, may open new perspectives for their optimization. Understanding the transition to turbulence and the associated enhancement of heat and mass transfer is particularly critical for advanced engineering applications, such as compact chemical reformers and high-performance heat exchangers.

In this section, a comprehensive analysis of the hydraulic performance of TPMS structures in compact reformers was analyzed and systematized. It is revealing that topology is the dominant factor that governs the Fanning friction factor. Among the topologies studied (Gyroid, Diamond, Schwarz P), the Schwarz P structure consistently emerged as the lowest-friction option. Furthermore, increased voidage and larger unit-cell sizes reduce pressure drop, whereas smaller cells and thicker walls increase it. The studies conclude that traditional porous-media correlations, such as the Ergun and Kozeny-Carman equations, rapidly fail for these complex geometries, necessitating the development of topology-specific models. This behavior is attributed to the unique flow characteristics of TPMS. Even at low *Re*, these structures disrupt developing boundary layers. At high *Re*, they generate intense turbulence and elevated turbulent kinetic energy.

5. Heat transfer

In essence, compact reformers for endothermic chemical processes based on TPMS structures operate as highly integrated catalytic chemical reactors. Their primary objective is to efficiently transfer thermal energy from a hot fluid stream to a chemically reacting flow through catalyst-coated walls. The critical importance of heat transfer in this design comes directly from the demands of endothermic reactions, such as steam methane reforming. These processes are fundamentally limited by the rate at which thermal energy can be delivered to the active catalytic sites to drive the reaction forward. TPMS architectures, with their continuous, interpenetrating channels and high surface-to-volume ratios, provide an ideal framework for process intensification. They enable rapid and efficient heat delivery directly to the catalyst-laden surfaces, thereby minimizing radial thermal gradients and preventing the formation of local cold spots that quench reaction rates and diminish conversion. Consequently, optimizing the heat transfer performance of a TPMS reactor is not merely advantageous but essential for achieving the high volumetric productivity and compactness required in next-generation chemical reformers.

5.1. Convective heat transfer

In the preceding sections, it was demonstrated that the complex geometry of TPMS structures disrupts fluid flow, thus improving heat transfer, and that their mathematical nature allows performance to be optimized through specific design parameters [113–115]. However, only a few studies have evaluated their advantages over traditional heat exchangers used in convection applications [116–118]. Therefore, this subsection reviews recent studies on convective heat transfer for different TPMS structures.

First, Qin et al. investigated the convective heat transfer performance of a Gyroid TPMS structure based on mathematical function control [119]. The authors proposed a factor, β_w, to quantify surface "wrinkle" formation and its role in enhancing convective heat transfer performance. Heat transfer performance and flow characteristics were analyzed using both numerical and experimental methods. Experiments determined that β_w enhances the convective heat transfer coefficient by 4.3%–8.2% and the Nusselt number (Nu) by 0.7%–3.5%. It was also proposed that a coefficient of $\beta_w = 0.8$ achieves optimal results. These results clearly show that the Gyroid TPMS structure is suitable for compact solutions due to its high convective heat transfer coefficient and Nu in small-scale heat exchangers.

To compare the thermal performance of the Gyroid structure in the aforementioned paper with another TPMS type, Wei et al. proposed a new three-fluid heat exchanger based on Diamond and Gyroid TPMS structures [120]. Numerical experiments were conducted to compare the heat transfer performance between the Diamond and Gyroid structures. The CFD results clearly showed that the three-fluid heat exchanger enables efficient heat exchange between each flow channel and air. When the solid volume fraction of the TPMS increases from 7% to 13%, Nu and the friction factor (f) on the cold side of the Diamond structure increases by 34% and 19.6%, respectively; meanwhile, for the Gyroid structure, they increases by 13.2% and 5.9%. Consequently, the Diamond TPMS structure was proposed as a compact heat exchanger solution due to its superior thermal performance compared to Gyroid structure.

The previous papers compared different TPMS structures with each other. However, a comparison between standard and TPMS heat exchangers was lacking. To address this gap, Zhang et al. proposed a new conformal filling method for specially shaped TPMS heat exchangers [121]. This study analyzed and investigated different TPMS structures based on tubes, Schwarz D, I-WP, Schwarz P, and Gyroid using both numerical simulations and experiments. The experimental results demonstrated that TPMS structures improve heat transfer efficiency in comparison to conventional heat exchangers. In particular, while the pressure drop in the straight-tube configuration is lower, the Nu of the straight-tube heat exchanger is 63.4% to 95.8% lower than that of the Gyroid structure. Furthermore, the TPMS-based heat exchanger provided a larger heat transfer area compared to straight tubes. Additionally, the Primitive structure exhibited significant performance improvement at low Reynolds numbers (Re). These results clearly show the advantages of TPMS structures over conventional tubes and highlight the performance variations between different TPMS types. Overall, this research not only demonstrates the superior performance of TPMS structures over conventional heat exchangers but also provides insights into the optimal TPMS structure based on the flow regime.

The studies discussed above focus mainly on the dependencies of Re, f, and Nu. However, in the flow dynamics section, it was determined that TPMS structures break the thermal boundary layers. Consequently, to effectively utilize TPMS structures in reformer applications, it is necessary to establish a correlation between Nu and the Prandtl number (Pr). To this end, Guillermo et al. investigated heat transfer and fluid flow through a Gyroid TPMS structure for a wide range of working fluids in the laminar regime [122]. CFD modeling was implemented to compute heat transfer parameters such as temperature (T), Nu, and Pr in TPMS structures. The authors found that changing Pr from 0.7 to 3 significantly reduces the fluid temperature, while a change from 10 to 14 results in a smaller temperature difference. Furthermore, a correlation for Nu as a function of Pr and Re was proposed for the Gyroid structure, expressed as:

$$Nu = 0.8831 \cdot Re^{0.5325} \cdot Pr^{0.4273}. \tag{5}$$

This correlation is valid within the ranges $10 < Re < 300$ and $0.7 < Pr < 14$. Insights from this paper allow for the straightforward calculation of heat transfer under forced convection for different flow conditions and working fluids in a Gyroid TPMS-based structure.

These studies provide valuable insights into the application of TPMS structures in heat exchangers. However, most existing research focuses mainly on evaluating the heat transfer coefficient (HTC) and the Nusselt number (Nu). For a comprehensive understanding of thermal performance, both effective thermal conductivity and HTC must be considered, as their combined effect ultimately dictates practical efficiency.

A summary of heat transfer performance is presented in Fig. 9, which illustrates the relationship between Nu and Re for different TPMS configurations. Fig. 9(a) indicates that in low-Reynolds number flows, the Nu-Re relationships exhibit similar characteristics for Diamond, Gyroid, and Primitive structures. In contrast, Fig. 9(b) shows that in high-Reynolds number flows, these relationships diverge significantly for all structures studied.

Therefore, the following section presents an evaluation of thermal performance based on the overall heat transfer coefficient, which comprehensively accounts for these key parameters.

5.2. Thermal performance evaluation

As demonstrated previously, parameters such as the Nusselt number (Nu), the Reynolds number (Re), thermal conductivity (k), and the heat transfer coefficient (h) collectively determine the practical applicability of Triply Periodic Minimal Surface (TPMS) structures in chemical engineering processes. Furthermore, these parameters are integral components of the overall heat transfer coefficient (OHTC). Therefore, in order to effectively utilize TPMS structures as compact solutions for chemical reformers, their OHTC must be thoroughly evaluated. This subsection systematically evaluates key thermal characteristics, including the OHTC and temperature distribution.

First, the recent work by Chen et al. evaluated the thermal performance of additively manufactured heat sinks based on Gyroid TPMS structures [123]. The study compared two types of Gyroid structures with a traditional plate-fin heat sink using both numerical simulations and experiments. The results demonstrated that the modified Gyroid structures exhibit a significantly lower surface temperature rise and more uniform thermal gradients than the plate-fin design. The thermal

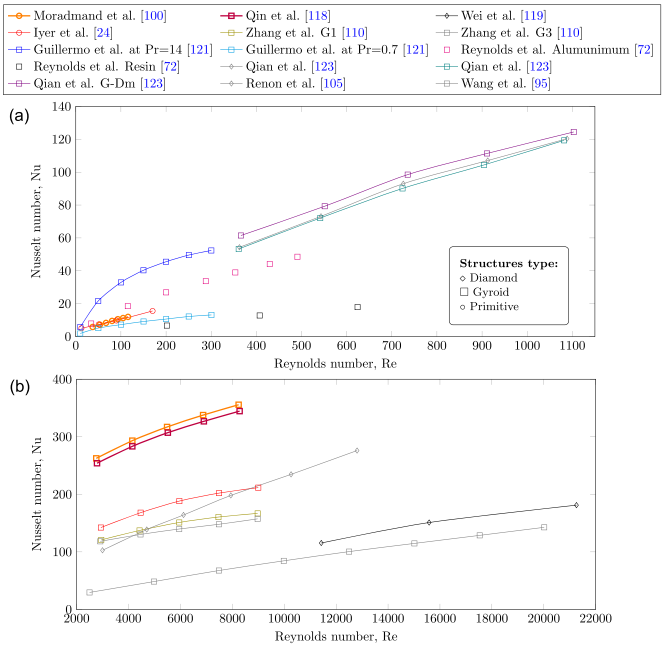


Fig. 9. Nu as a function of Re: line – numerical simulations, points – measured data. (a) Low Re and Nu; (b) high Re and Nu. ⋄ – Diamond; ⬠ – Gyroid; ∘ – Primitive.

efficiencies for the Gyroid-based and plate-fin structures were determined to be in the range of 16.7%–64.8% and 7.2%–29.4%, respectively, at flow rates of 1 to 10L/min. Moreover, at the maximum flow rate (10L/min), the plate-fin heat sink exhibited at least four times more thermal resistance than the Gyroid structure. These findings highlight that optimizing TPMS structures can significantly enhance thermal performance in heat exchanger applications. However, the study did not investigate the influence of different construction materials.

To address this gap, Reynolds et al. investigated the thermal performance and overall heat transfer coefficient of additively manufactured heat exchangers, focusing on the influence of material selection and design optimization [73]. The study analyzed TPMS heat exchangers fabricated from various materials, including green and sintered alumina, nylon, aluminum, and resin. Structural properties such as voidage, hydraulic diameter, and wall thickness were characterized by CT to determine their impact on heat transfer efficiency. Experimental results demonstrated that aluminum-based heat exchangers outperform alternatives, achieving higher OHTC values. Specifically, aluminum outperformed the resin-based TPMS heat exchanger by a factor of 1.05 to 1.16. Furthermore, at Reynolds numbers exceeding 400, the OHTC for the resin heat exchanger decreased significantly compared to those of aluminum and sintered alumina. The article also addresses trade-offs, noting that while aluminum improves thermal performance, it introduces manufacturing challenges and increased costs. The authors emphasized that without topological analysis, the results for each material would have deviated from the experimental data, with systematic errors in the Nusselt number of up to 20%. Consequently, they highlighted the need for a new Nusselt number correlation specific to TPMS structures.

To resolve this issue, Qian et al. conducted a comprehensive investigation into the optimization design and heat transfer performance of TPMS-based compact heat exchangers using the field synergy principle [124]. The study analyzed Gyroid, Diamond, and a novel hybrid G-D structure through combined CFD simulations and experimental validation. The flow characteristics and thermal performance of the G-D heat exchanger were compared with those of conventional TPMS structures. Based on the experimental results, new correlations for Nu and the

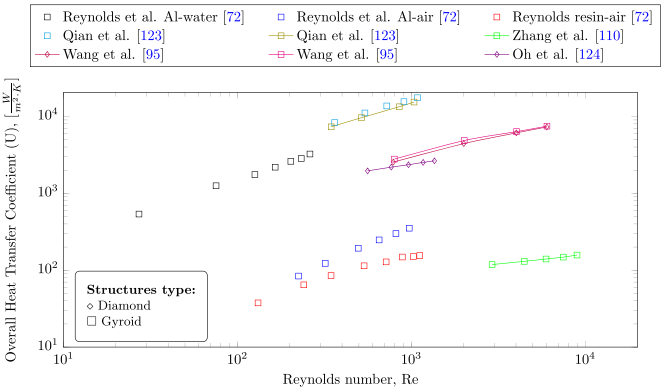


Fig. 10. Overall heat transfer coefficient as a function of Re: line – numerical simulations, points – measured data.

friction factor (f) with respect to Re were proposed for the G-D heat exchanger:

$$Nu_{G-D} = 3.748 \cdot Re^{0.420} \cdot Pr^{1/3}; \quad f_{G-D} = 2.815 \cdot Re^{-0.188}. \tag{6}$$

The results demonstrated that the G-D hybrid configuration achieves significantly enhanced heat transfer efficiency and improved compactness relative to both traditional heat exchangers and standard TPMS designs. Specifically, at low Re, the heat transfer per unit volume for the G-D structure increases by up to 31.3% compared to the Gyroid and by up to 11% compared to the Diamond structure. Consequently, the authors concluded that the G-D TPMS structure shows significant potential for heat exchanger applications.

The investigations reviewed here evaluated the overall heat transfer coefficient and thermal performance of TPMS-based heat exchangers. The results consistently demonstrate that TPMS architectures provide a compact and efficient solution, achieving significantly higher OHTC values than conventional designs. As illustrated in Fig. 10, which compiles data from recent studies and foundational works, the OHTC exhibits a strong dependence on Re across all TPMS structures, with the Gyroid-type showing particularly promising results.

The demonstrated thermal advantages, combined with the inherent compactness, make TPMS heat exchangers a promising solution for many industries [125]. Energy conversion systems can utilize their improved heat recovery capabilities, while chemical process equipment can benefit from their unique structural characteristics, which enhance both heat and mass transfer. This dual enhancement is particularly valuable for applications that involve chemical reactions.

In this section, the systematic evaluation of the thermal behavior of TPMS structures implemented in heat exchangers is evaluated. Convective heat transfer studies show that TPMS structures cause pronounced boundary-layer disruption and generate secondary flows. These mechanisms elevate Nu far beyond the predictions of conventional correlations. For instance, Gyroid and Diamond topologies achieve Nu of 63–96% higher than those in straight tubes. Additionally, optimized aluminum TPMS structures achieve overall heat transfer coefficients that deliver lower thermal resistance than plate-fin designs at comparable pressure drops. However, manufacturing defects and material choices can introduce 11–48% performance variability.

Finally, the overall thermal performance is strictly governed by the coupling of convective heat transfer in the fluid domain and the effective thermal conductivity. Although tortuous fluid pathways significantly elevate the convective heat transfer coefficient, this high convective efficiency can only be fully exploited if the solid structure of the TPMS possesses sufficient thermal conductivity to distribute heat throughout the volume. Because TPMS structures act as porous media, their k_{eff} is fundamentally limited by both the bulk material's thermal conductivity

and the topology's voidage. As the voidage increases, the cross-sectional area available for solid conduction decreases, leading to a higher intrinsic thermal resistance. Consequently, in systems combining high fluid velocities with high voidage, thermal conduction through TPMS becomes the rate-limiting step. Voidage and unit-cell size emerge as dominant design parameters: increasing voidage trades are conductive for convective efficiency, while smaller cells amplify heat transfer intensity.

6. Catalysis & reformers

As previously described, TPMS-based structures are a promising solution for chemical reactors as a result of their high surface-to-volume ratio, compactness, and lightweight nature. In general, chemical reactors for endothermic catalytic processes function also as heat exchangers: heat is transferred from a hot flow to a chemically reacting flow through a wall, with one side of the wall coated with the catalyst. Analyzes of flow dynamics and heat transfer confirm that TPMS structures significantly enhance thermal performance. This section focuses on the subsequent step in TPMS analysis: evaluating their behavior as catalyst-coated carriers for chemical reactors facilitating endothermic processes.

Conventionally, catalysts in chemical engineering processes are employed as simple shapes such as spheres, cylinders, or Raschig rings [126]. These are typically porous materials [127]. A significant drawback of such configurations is the diffusion limitation. The porous structure of catalyst particles impedes mass transfer, and the reaction rate decreases towards the particle's interior as the reactant molar fraction decreases and the product molar fraction increases [89]. Consequently, a portion of the catalyst material remains underutilized. Addressing this diffusion limitation is therefore critical to improving catalytic efficiency.

The accurate prediction of reaction behavior also relies on robust kinetic models. For steam methane reforming, two widely cited models are those developed by Xu and Froment [128] for a Ni-based catalyst and by Hou and Hughes [129] for a Ni/Al_2O_3 catalyst. Both models were derived using the Langmuir–Hinshelwood–Hougen–Watson (LHHW) technique and share similar mathematical forms.

In contrast, the ammonia decomposition process is often described by the Temkin-Pyzhev kinetic model [130]. Originally developed for iron catalysts, this model is based on the concept of non-uniform catalyst surfaces and assumes that the associative desorption of nitrogen atoms is the rate-determining step. Thus, its mathematical form differs significantly from the LHHW-based models.

A common limitation of these kinetic models (LHHW and Temkin-Pyzhev) is that they neglect the influence of catalyst particle size. As a result, they cannot account for the internal diffusion limitations. Increasing the size (thickness) of catalyst particles does not yield a proportional increase in the product yield or conversion. Beyond a certain size, the internal diffusion resistance becomes the dominant factor, rendering additional catalyst depth ineffective.

Utilization of TPMS structures as catalyst carriers enables the solution of these problems. The high surface-to-volume ratio allows the application of a thin, wash-coated catalyst layer, making all of the active material accessible to the reactants. This can be achieved via various methods, such as dip coating and spin coating. This high catalyst utilization directly translates into significant economic and size efficiency, reducing the total amount of catalyst required compared to traditional packed beds.

This approach opens new avenues for implementation in thermal and chemical engineering. One promising application is in thermochemical exhaust gas heat recovery systems, where compactness and enhanced performance are crucial. Another development is the efficient, compact reactor based on TPMS structures for hydrogen storage. To understand the current state of development and the implementation challenges for these compact TPMS reformers, the following section reviews contemporary research.

6.1. TPMS for hydrogen storage

The advantages of TPMS structures, as previously outlined, create new possibilities in chemical engineering. As structured catalyst supports, TPMS can significantly enhance heat and mass transfer within reactors due to their superior geometrical characteristics. These characteristics are crucial for improving performance in various applications, including hydrogen storage reactors, mobility systems, and electrochemical and conventional endothermic reactors. This subsection describes several TPMS-based reactors for such chemical applications.

First, for hydrogen storage, Jiang et al. recently investigated the synergetic enhancement of hydrogen concentration and heat transfer in a TPMS-based reactor [131]. Their study involved a comprehensive CFD analysis of multi-physical interactions, including heat transfer, chemical reactions, and stress. The results demonstrate that reactors employing TPMS structures exhibit superior heat transfer and hydrogen storage performance compared to those using traditional fins, as illustrated in Fig. 11. The authors also noted that the region near the wall maintains a temperature lower than the interior, creating favorable conditions for hydrogen absorption. These findings indicate the strong potential of TPMS structures as catalyst carriers in hydrogen storage applications.

Also focusing on hydrogen storage, Abe et al. explored the potential of TPMS structures for high-performance hydrogen storage in mobility applications [132]. This study numerically simulated the mechanical performance and hydrogen absorption/desorption characteristics of a metal hydride reactor. Several TPMS structures - Diamond, Gyroid, Schwarz-P, and Lidinoid - with varying geometrical properties were evaluated. The results, shown in Fig. 12, indicated that the Lidinoid structure achieved the shortest completion times for both absorption and desorption processes. Fig. 12 depicts the time evolution of hydrogen concentration during desorption for the different TPMS structures. These results demonstrate a clear dependence of absorption/desorption performance on the surface-to-volume ratio of the TPMS, as summarized in Table 2. The combination of high heat transfer performance, rapid hydrogen absorption/desorption, and excellent mechanical properties makes TPMS structures well-suited for electrochemical reactors and other chemical applications.

Extending the application of TPMS to solid-state hydrogen storage, Lin et al. [133] conducted numerical modeling to investigate the thermal management and absorption kinetics of TPMS-integrated reactors. The authors conducted numerical simulations for the Gyroid structure in comparison with traditional ribbed designs and alternative TPMS structures, in particular, Diamond and Primitive. The computational results demonstrated that at a voidage of 0.8, the Gyroid-based structure achieved 90% hydrogen absorption in 183 s, significantly outperforming the conventional fin structure's 273 s while maintaining a lower average temperature variation rate of $0.664\,K\,s^{-1}$. Furthermore, compared with a voidage of 0.6, the Gyroid structure consistently shows an improved heat transfer efficiency, producing a temperature change rate of 0.614 compared to $0.742\,K\,s^{-1}$ for the Diamond and $0.638\,K\,s^{-1}$ for the Primitive structures. Thus, the study clearly concludes that the Gyroid structure has an improved geometric structure for optimizing heat and mass transfer in the hydrogen storage application.

Enhancing the cost-competitiveness of these technologies requires designing reactors with higher power density, for which optimization of TPMS-based reactors is essential. In this context, Barzegari and Forner-Cuenca investigated the topology optimization of porous electrodes for electrochemical flow reactors using TPMS structures [134]. Using numerical methods, they modeled the electrode as both 2D and 3D TPMS. In all simulated cases, the pump power loss was reduced by 61–98%, depending on the operating conditions and the Darcy number. Regarding electrochemical performance, an improvement of up to 29%

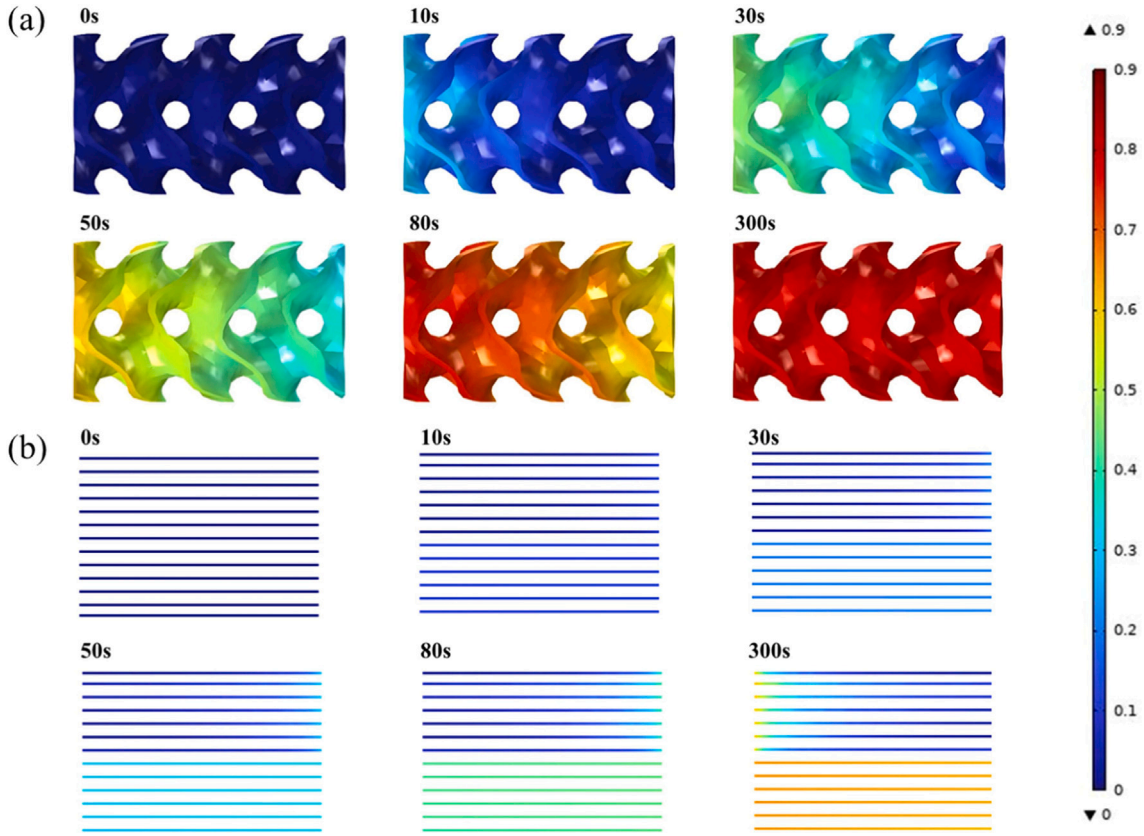


Fig. 11. Hydrogen concentration distribution at different time steps for the following cases: (a) TPMS hydrogen storage reactor; (b) a finned reactor operating at a pressure of 1.49 MPa and an initial temperature of 293.15 K [131].

was observed; however, in some configurations, performance decreased by up to 12% due to the formation of low hydraulic resistance bypass channels. The authors note that their study did not account for detailed electrolyte-electrode interface effects, localized fluid flow phenomena, or mass transfer. Consequently, the actual performance may vary, necessitating further simulation and experimental validation to fully evaluate TPMS behavior.

TPMS structures also show promise for micro thermophotovoltaic (TPV) systems. Pioneering work by Gao et al. investigated enhancing micro-TPV system efficiency using a Gyroid lattice-embedded micro combustor [135]. Using CFD, they compared the performance of a Gyroid-based microcombustor with those featuring aligned and staggered pin-fin arrays. The results clearly show that the TPMS combustor achieved the highest system efficiency among the three configurations with the same input velocity. Specifically, at an inlet velocity of 25 m/s, the system efficiency with the TPMS reached 6.19%, which is 9.6% and 4.2% higher than the aligned and staggered pin-fin arrays, respectively. This demonstrates that the implementation of TPMS structures in micro combustors can significantly enhance thermal performance.

As shown in the above studies, TPMS structures are widely applicable across different chemical engineering domains. In these applications, they typically enhance both compactness and thermal efficiency, making them excellent candidates for use in compact reformers. Therefore, based on this evidence, TPMS-based structures hold significant promise for application in chemical reformers.

6.2. *TPMS for H_2 production via fuel reforming*

Current research on TPMS-based structures for endothermic chemical reactions primarily focuses on fuel reforming to produce hydrogen-rich gas; a key objective is to compare the conversion efficiency and flow dynamics of different TPMS architectures, aiming to improve reactor efficiency and reduce its footprint. The potential advantages of this technology make it a compelling area of investigation. The following section provides an overview of scientific studies that employ TPMS as a catalyst support for chemical reactions.

Iwaniszyn et al. experimentally and numerically evaluated the flow and heat transfer performance of 3D-printed Gyroid TPMS structures as catalyst carriers [74]. Their methodology combined CFD simulations with physical experiments on structures fabricated via selective laser melting from stainless steel. The authors designed structures with a consistent voidage of 81% but varying unit cell sizes to assess the impact on thermal performance. Using air as the working fluid, they demonstrated that increasing the cell diameter from 2 to 4 mm resulted in a doubling of both the heat transfer coefficient and the pressure drop. Based on these findings, new pressure drop and heat transfer correlations specifically for TPMS geometries were developed, expressed as:

$$f = 3.607 \cdot Re^{-0.206}; \quad Nu = 0.38 Re^{0.745} Pr^{0.33}. \tag{7}$$

The study also compared traditional catalytic supports, such as foams and packed beds, with the TPMS structures. The results indicated that the TPMS designs demonstrated thermal efficiency equivalent to these traditional supports. Therefore, TPMS structures represent a high-performance alternative that offers enhanced heat and mass transfer for chemical applications.

In a chemical application, Italiano et al. experimentally evaluated the performance of 3D-printed Ni-alloy TPMS structures, activated with a Ru/Al_2O_3 catalyst, for carbon-free hydrogen production via ammonia decomposition [31]. The study investigated structures based on BCC, Kelvin, and Gyroid unit cells, fabricated via laser powder bed fusion with a consistent cell size but varying strut dimensions. The Ni-alloy supports were prepared for catalytic use by depositing a 5 wt.% Ru/Al_2O_3 catalyst through a dip-spin coating technique. Experimental

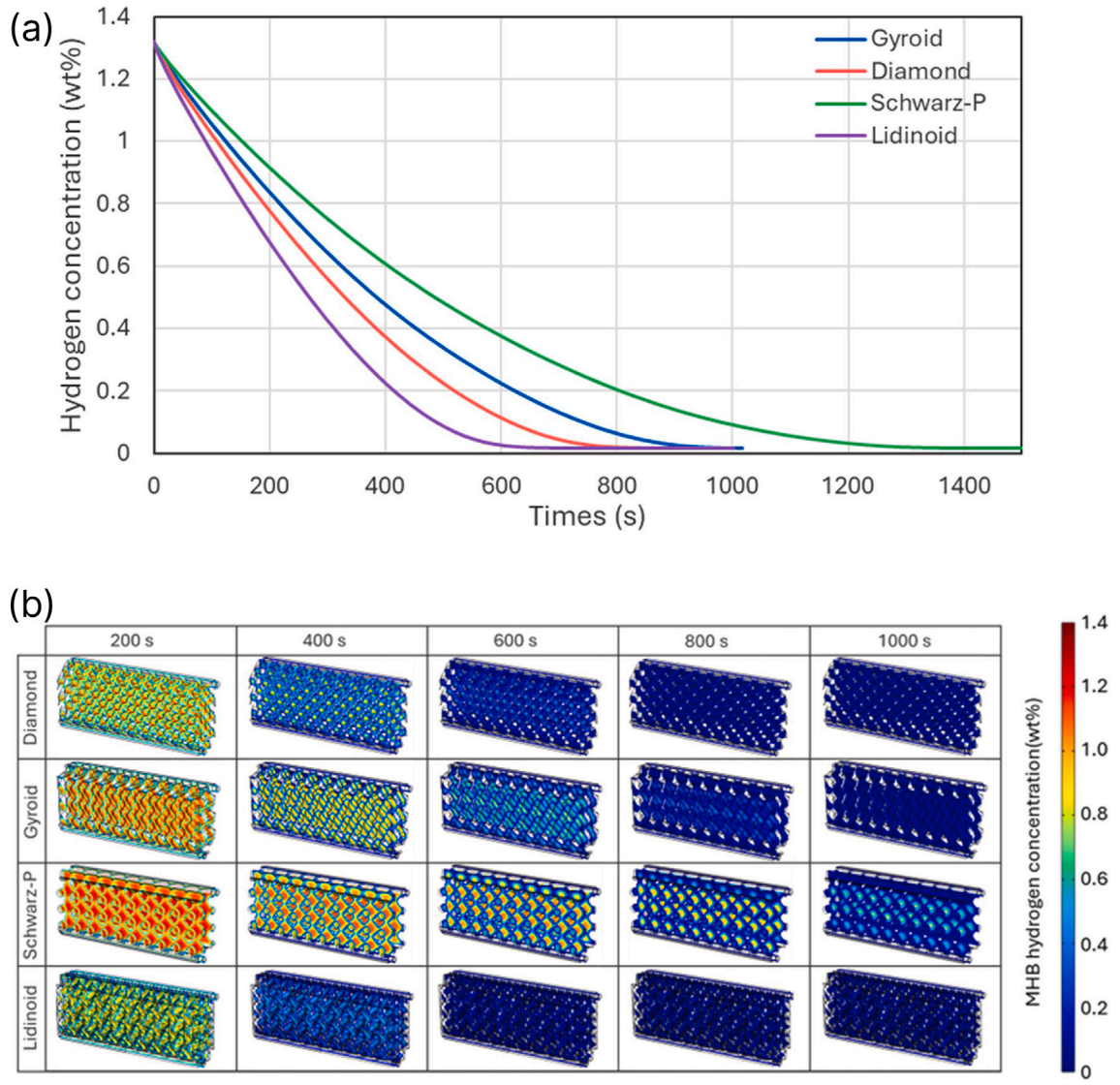


Fig. 12. Discharging profile (a) and metal hydride bed (MHB) hydrogen concentration distribution (b) for different TPMS models during hydrogen desorption [132].

results demonstrated that the Gyroid structure exhibited superior performance, particularly at lower temperatures (400–500 °C), achieving an ammonia conversion of 75% at 450 °C. Under identical conditions, the BCC and Kelvin structures recorded lower conversion rates of 71.3% and 64.4%, respectively. The Gyroid's intricate, tortuous flow channels were identified as the key factors for enhancing mass transfer and reactant diffusion. This internal geometry improves catalytic efficiency despite the uniform catalyst loading across all structures. Furthermore, the study noted that larger strut sizes in the BCC and Kelvin structures resulted in thicker catalytic layers, creating a compromise between catalyst utilization and diffusion limitations. These findings underscore the potential of the Gyroid TPMS structure for optimizing reactor designs for distributed hydrogen production, enabling high efficiency with reduced catalyst and reactor size requirements.

Building on this work, Richard et al. modeled the feasibility of integrating membranes into periodic open cellular structures for the ammonia decomposition process [136]. Using a combined numerical and experimental approach (Figs. 13 and 14), the study evaluated the influence of packing type and voidage on reactor performance. The numerical simulations highlighted these effects, as shown in Fig. 15. This figure indicates that the performance order at low superficial velocities is: Kelvin 3-0.6 with baffles > Kelvin 3-0.6 (without baffles) > empty bed > pellet bed. Furthermore, a trade-off between voidage and hydrogen permeation was observed, detailed in Fig. 15. The permeation increases with the voidage, reaching a maximum at a voidage of 0.8; however, further increases lead to a decrease in total hydrogen production due to kinetic limitations. Although this study focused on ammonia decomposition, the demonstrated benefits—namely improved transport properties and design flexibility—are transferable to other processes constrained by heat and mass transfer, such as methanol steam reforming.

Furthermore, Chen et al. experimentally evaluated the influence of TPMS structures as catalyst carriers on hydrogen production via methanol steam reforming (MSR) in micro-reactors [137]. The study focused on four TPMS configurations—Schwarz D, G, P, and I-WP—fabricated via 3D printing and coated with a Cu/Zn/Al/Zr catalyst using a sol-gel method. The results demonstrated that the Schwarz D structure outperformed the others, achieving the highest hydrogen production and methanol conversion rates, particularly at lower temperatures and higher flow rates. For instance, at a gas hourly space velocity of 17,065 mL g^{-1} h^{-1} and 250 °C, the D structure achieved a hydrogen flow rate of 0.02 mol h^{-1}, surpassing the P structure's conversion efficiency by 22% under identical conditions. Additional analyzes of pressure drop and catalyst stability revealed that the D and G structures offered superior heat transfer and lower pressure losses, thereby

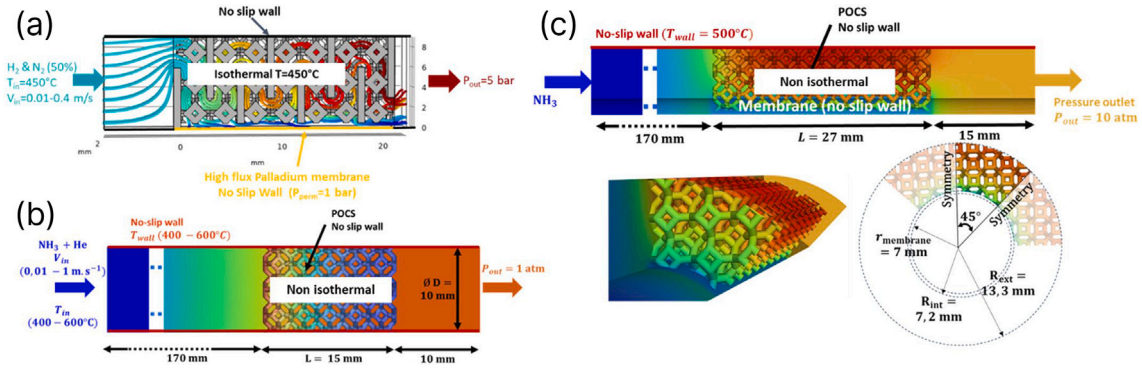


Fig. 13. Graphical summary of the numerical and experimental results from Richard et al. [136]: (a) implemented boundary conditions and physical setups; (b) permeation domain setup; (c) traditional reactor mirroring the experimental one, used to validate the chemistry.

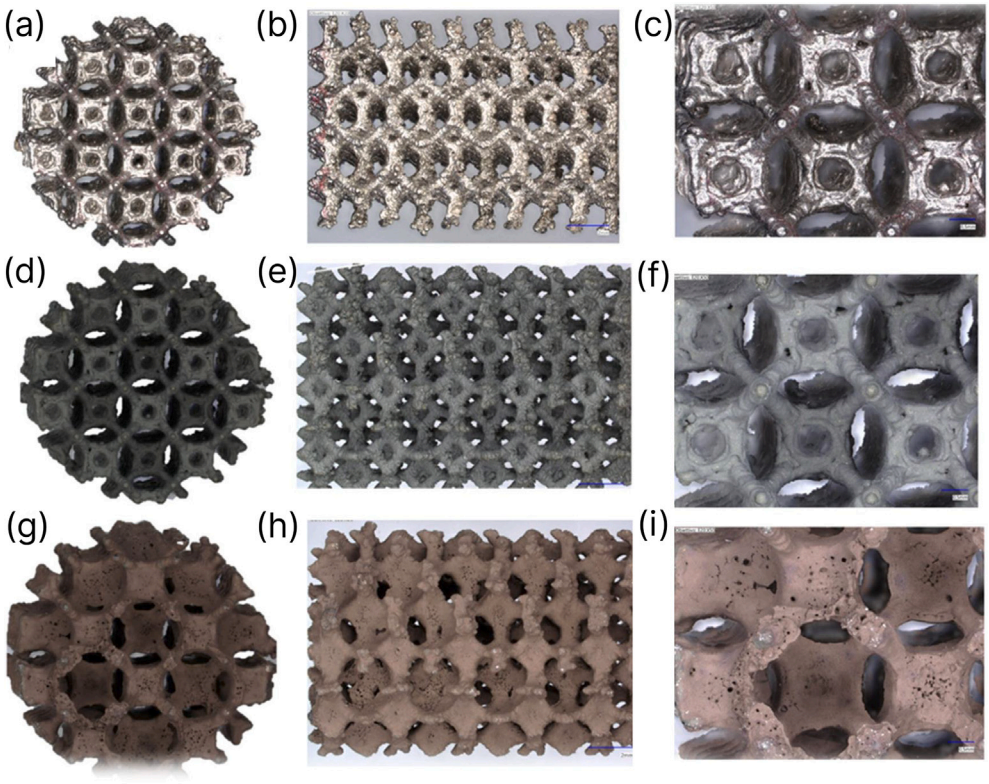


Fig. 14. Optical microscopic images of (a)–(c) Fresh Kelvin, (d)–(f) after 900 °C for 6 h, and (g)–(i) after washcoating [136].

enhancing reactor performance. In contrast, the P and I-WP structures exhibited issues with catalyst crowding and nonuniform flow, which impaired mass transfer. These results underscore the critical role of the carrier geometry, demonstrating that the Schwarz D structure provides an optimal balance between catalytic performance and operational stability.

In addition, for MSR, Li et al. [138] conducted a multiobjective optimization of 3D-printed structured catalysts and reactors for high-efficiency vehicular applications. In this study, a novel hybrid TPMS (H-TPMS) structured catalytic reactor was designed by integrating common TPMS structures, specifically the Gyroid, Diamond, and Primitive. The authors conducted CFD simulations and revealed a positive correlation between the methanol conversion rate and the CO selectivity of the new H-TPMS SCR. Under constant temperature parameters, structures dominated by the Schwarz-D and Gyroid topologies exhibited high methanol conversion rates. However, they underperformed compared to the Schwarz-P dominated structures in minimizing pressure drop. The global optimum point on the three-objective Pareto frontier achieved a conversion rate of 97.8%, a CO selectivity of 1.78%, and a pressure drop of 14.6 Pa, which are parameters of a uniform H-TPMS structure. Also, comparative CFD analyzes demonstrated that the intricate, interconnected spiral channels within the hybrid structure significantly improved gas transport and diffusion in chemical reactor. In conclusion, these computational results show that combining different TPMS structures is a mathematical approach to balancing the trade-offs of different structures.

Building on this, Liu et al. conducted a numerical investigation of coated TPMS structures for hydrogen production via MSR [139]. Using CFD, the study assessed fluid flow and reaction kinetics for three TPMS structures—Schwarz P, G, and D—across a voidage range of 50 to 90%. The simulations indicated that the curved surface of the Schwarz D structure yielded the highest reaction efficiency. At 50% voidage, it produced

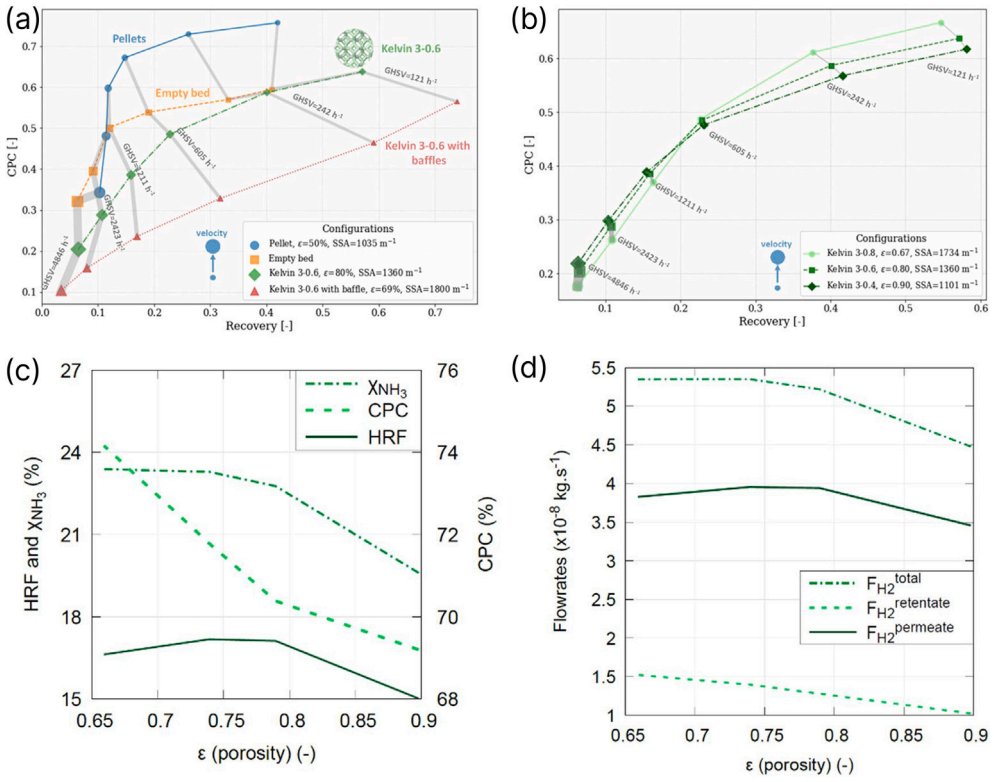


Fig. 15. Numerical simulation results showing the influence of the packing type on permeation performance at T = 500 °C, P_{ret} = 5 bar, P_{perm} = 0.1 bar, and 50% H_2 inlet: (a) – performance comparison across several configurations (pellets with 3 mm diameter, Kelvin 3-0.6, empty bed); (b) – impact of porosity on the Kelvin cell type; (c) – NH_3 conversion and hydrogen recovery; (d) – hydrogen productivity for different porosity values of the KC lattice [136].

a hydrogen concentration more than 2.5 times higher than that of the P structure and 1.5 times greater than that of the G structure, which is attributed to its larger specific surface area and complex flow paths. Conversely, the Schwarz P structure exhibited the lowest pressure drop (0.15 Pa) and the highest permeability (6.31×10^{-9} m^2) at 90% voidage, while the G structure maintained good flow performance throughout the entire 50–90% porosity range. At 70% voidage, the hydrogen concentrations for all three structures were comparable, with values of approximately 0.09, 0.13 and 0.16 mol m^{-3} for the P, G, and D structures, respectively. These findings illustrate the distinct advantages of different TPMS geometries: the Schwarz D structure enhances the reaction efficiency, whereas the P and G structures are more beneficial under specific conditions requiring low flow resistance.

Extending methanol steam reforming (MSR) reaction study, Li et al. [140] evaluated the efficacy of TPMS structured micro-reactors for hydrogen production. The authors conducted numerical simulations where bio-inspired TPMS topologies were compared specifically against Schwarz (S-type) and Diamond (D-type) structures against conventional micro-pin reactors to investigate the impact of structural design and operating parameters. The computational simulations indicated that the TPMS-based structures consistently outperformed conventional designs due to their interconnected voidage networks and high specific surface areas. In particular, the numerical data revealed a distinct geometric trade-off: a 60% voidage has improved hydrogen production kinetics, whereas an 80% voidage maximized thermal management. Additionally, increasing the gas hourly space velocity (GHSV) and temperature improve heat transfer and reaction rates, raising hydrogen yield while suppressing CO selectivity, with optimal performance stabilizing at a steam-to-carbon ratio of 1.3. Finally, the computational model predicts that this optimized micro-reactor with a gradient TPMS structure can produce 5000 mL min^{-1} of hydrogen. This corresponds to a volumetric energy density of 17.56 kW L^{-1}, which is theoretically sufficient to power a 500 W fuel cell. These results highlight the potential of gradient TPMS structures for compact energy conversion.

In a related development, Asmat et al. [141] investigated an electrified reformer featuring bespoke, catalyst-coated additively manufactured monoliths with 3D pore architectures for syngas production, as shown in Fig. 16. This study combined CFD and experimental methods to compare a traditional hexagonal honeycomb monolith with TPMS structures, including Gyroid, Octet, and Voronoi lattices, under induction heating (Fig. 17). The results demonstrated that the Gyroid structure promoted smooth velocity streamlines, tortuous flow paths, high particle residence time, and a low pressure drop. In contrast, the Octet and Voronoi structures exhibited more random geometries, higher pressure drops, and rougher velocity streamlines, which adversely affected fluid mixing and catalyst activity. The Gyroid-based reactor achieved > 99% CH_4 conversion at 900 °C and remained stable for > 42 hours without evidence of coke deposition. Meanwhile, reactors based on the Octet and Voronoi structures showed slightly lower conversion rates, with pressure drops 6 and 4 times higher, respectively. Overall, these findings highlight the significant potential of 3D-printed TPMS-based structures, particularly when integrated with induction heating, not only for dry methane reforming but also for various other catalytic reactions.

Continuing the focus on steam methane reforming, Bezerra et al. [142] investigated the application of 3D-printed metallic monoliths featuring TPMS structures as advanced supports for dry and steam methane reforming. To evaluate structural integrity and resistance to carbon deactivation under severe thermal conditions (700–800 °C), the authors fabricated the Inconel 718 Gyroid-based structure using SLM and applied various coating methods (W_1, W_2 and W_3) to coat the micro-channels with the Ni/Al_2O_3 active phase as shown in Fig. 18 after steam methane reforming. In comparison, with traditional cordierite monoliths, the experimental data showed that the specific deposition technique strongly influences the catalytic stability and dispersion of the

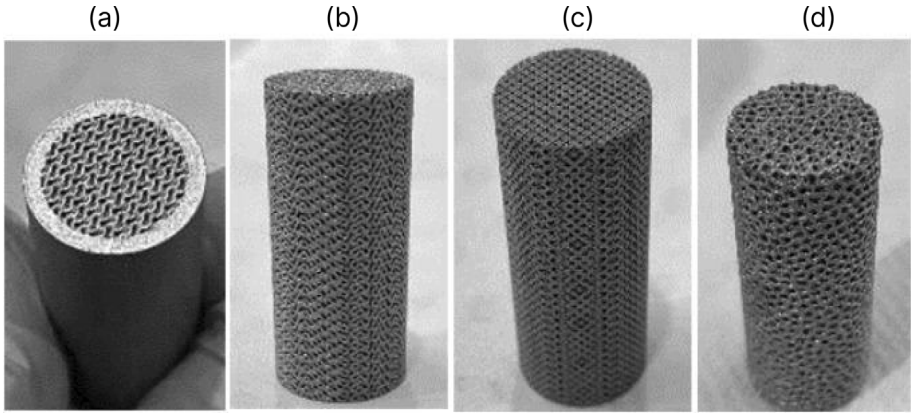


Fig. 16. Different printed TPMS-based structures (according to Asmat et al. study): (a) 3D-printed Gyroid monolith with wall (GW), (b) 3D-printed Gyroid monolith without wall (GWW), (c) 3D-printed Octet monolith without wall (OWW), and (d) 3D-printed Voronoi monolith without wall (VWW) [141].

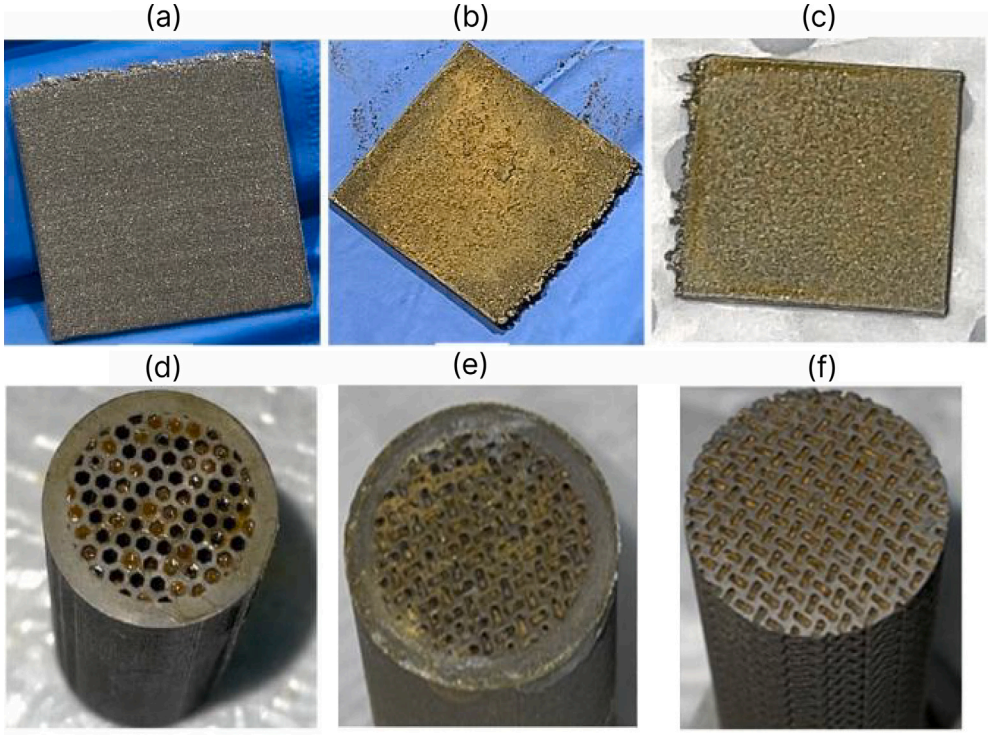


Fig. 17. (a) Uncoated 3D-printed flat plate, (b) Slurry-coated $NiO/Ce_{0.8}Gd_{0.2}O_{2-\delta}$, (c) Precursor solution-coated $NiO/Ce_{0.8}Gd_{0.2}O_{2-\delta}$ plate, (d) Precursor solution-coated $NiO/Ce_{0.8}Gd_{0.2}O_{2-\delta}$ honeycomb monolith, (e) Precursor solution-coated $NiO/Ce_{0.8}Gd_{0.2}O_{2-\delta}$ GW (gyroid walled) monolith, (f) Precursor solution-coated $NiO/Ce_{0.8}Gd_{0.2}O_{2-\delta}$ GWW (gyroid without wall) monolith. All plates and monoliths were made of Inconel-625 [141].

active phase. Specifically, at 700 °C under dry reforming, the W_3 method yielded a highly stable catalyst by promoting a Ni/Al_2O_3 phase that enhanced the dispersion of finer nickel particles. In contrast, the W_2 coated technique demonstrated superior overall activity and selectivity. Under steam reforming conditions, the reduction in carbon deposition allowed the active sites to remain active, resulting in a uniformly high level of methane conversion for all deposition methods. This conclusion is confirmed by the measured H_2/CO ratios in the synthesis gas. Finally, the study shows that the internal corrosion resistance of Inconel, combined with the thermal stability and efficient heat dissipation inherent in the Gyroid-based structures, provides a highly reliable basis for optimizing active phase deposition and scaling up industrial hydrogen production.

The comprehensive review of TPMS applications in chemical reformers demonstrates their strong potential as advanced catalyst supports. These structures exhibit superior heat and mass transfer characteristics compared to conventional configurations. However, their implementation faces several technical challenges, including the precise additive manufacturing of TPMS geometries and the uniform deposition of catalytic coatings on their complex surfaces. Although the studies mentioned above provide valuable insights, each focuses on distinct aspects of TPMS performance. Collectively, they underscore the need for large-scale integrated studies to fully elucidate the underlying processes and optimize these promising structures for industrial application.

In this section, TPMS structures were evaluated as catalyst supports in compact reformers for hydrogen production and storage. It was found that Gyroid and Diamond-based supports outperformed traditional bulk and ceramic foam catalysts by a factor of 2–10 in terms of volumetric reaction rate and by 50–90% in terms of pressure drop reduction,

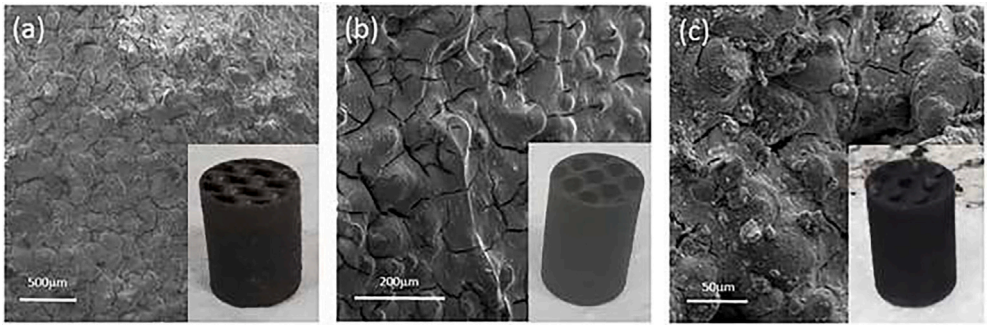


Fig. 18. SEM images after steam methane reforming: (a) $Ni/Al_2O_3-W_1/MM$; (b) $Ni/Al_2O_3-W_2/MM$; (c) $Ni/Al_2O_3-W_3/MM$ [142].

while achieving near-equilibrium conversions at significantly reduced temperatures. Also, across steam methane reforming, methanol steam reforming, ammonia decomposition, and dry/steam CO_2 reforming, TPMS reactors demonstrate 92–99% methane conversion at 650–750 °C (Ni/Rh catalysts), > 95% ammonia conversion at only 450–550 °C (Ru/Fe), near-100% methanol conversion at 300–350 °C, and 35–65% CO_2 conversion at 700–800 °C. These findings underscore the potential of TPMS-based catalytic reformers for on-board hydrogen production.

7. Challenges and future prospects

This review posits that structures based on Triply Periodic Minimal Surfaces (TPMS) represent a promising approach for developing compact chemical reformers, owing to the unique advantages detailed in Section 2. The advancement of such compact reformers is critical for two key applications: on-board hydrogen production [143,144] and thermochemical exhaust heat recuperation [92,145–147], where compactness and high efficiency are paramount for integration with internal combustion engines or gas turbines.

Thermochemical exhaust heat recuperation (TCR) is a technology that enhances the thermal and combustion efficiency of fuel-consuming equipment, such as internal combustion engines, gas turbines, and industrial furnaces [148–150]. The core principle of TCR involves utilizing waste exhaust heat to drive endothermic, pre-combustion reforming of the primary fuel [151]. In this process, the enthalpy of the exhaust gas is converted into chemical energy stored in the reformate – the product of fuel reforming. Consequently, the heat of combustion of the reformate is greater than that of the original primary fuel on a per-mole basis [152,153].

A significant barrier to implementing TCR systems in internal combustion engines and gas turbines is the stringent requirement for the reformer's compactness and effectiveness. Conventional thermochemical reformers are often tubular and packed with catalyst particles, a design that tends to be bulky [154]. The implementation of compact reformers with TPMS-based structures offers a solution, enabling not only high thermal efficiency but also unparalleled design flexibility. The innate cellular architecture of TPMS allows for the manufacture of reformers in virtually any required shape to fit spatial constraints.

Furthermore, the growing emphasis on hydrogen as a clean fuel has spurred interest in reformers for on-board hydrogen production [155]. In these systems, a primary fuel or hydrogen carrier, e.g., ammonia or a liquid organic hydrogen carrier (LOHC), is reformed to produce hydrogen or a hydrogen-rich gas [156,157]. For mobile and portable applications, the compactness and effectiveness of the reformer are crucial. The use of TPMS-based structures in such on-board hydrogen production systems presents significant potential for their widespread adoption in daily applications.

This review has demonstrated that Triply Periodic Minimal Surface (TPMS) structures represent a transformative approach for developing compact chemical reformers, offering exceptional heat and mass transfer characteristics, high surface-to-volume ratios, and design flexibility. However, the path from laboratory demonstrations to industrial implementation is obstructed by significant challenges that must be critically addressed. This section provides a balanced analysis of these obstacles and outlines future research directions.

7.1. Manufacturing and material challenges

Despite advances in additive manufacturing, fabricating TPMS reformers presents formidable difficulties. The complex internal channels characteristic of TPMS geometries create severe post-processing challenges. In powder-based methods such as selective laser melting, unsintered powder becomes trapped within cellular channels, and its complete removal is extremely difficult without destructive sectioning [64]. This trapped powder can subsequently be released during operation, contaminating catalysts or downstream components.

Furthermore, internal support structures required for overhanging features are nearly impossible to remove from finished parts, restricting design space to self-supporting orientations. Quality assurance presents another critical barrier: internal struts and walls cannot be inspected using conventional visual methods, forcing reliance on expensive and time-consuming micro-computed tomography for defect detection [65,66].

Material limitations compound these challenges. High-temperature alloys suitable for reforming reactions (Inconel, Hastelloy) are expensive and difficult to process. The high thermal gradients inherent in additive manufacturing induce residual stresses that cause warping or cracking, particularly in thin-walled sections that maximize surface area. Current printing speeds remain insufficient for mass production, limiting scalability to high-value, low-volume applications.

Future directions: Development of powder-free manufacturing methods; in-situ process monitoring for defect detection; design-for-additive-manufacturing guidelines specifically for TPMS reformers; exploration of lower-cost materials; and standardization of quality assurance protocols.

7.2. Catalytic coating and integration challenges

The application of uniform, adherent catalytic coatings on complex TPMS surfaces remains a critical unresolved challenge. While dip coating and spin coating have been demonstrated at laboratory scale, achieving thickness uniformity on internal surfaces with varying orientations is extraordinarily difficult. Gravitational drainage during coating creates thicker deposits on downward-facing surfaces and thin—or bare—regions on upward-facing surfaces [136].

The trade-off between coating thickness and catalytic effectiveness is particularly acute in TPMS structures. Thick coatings increase catalyst loading but introduce internal diffusion limitations—precisely the problem that TPMS architectures aim to solve. Thin coatings eliminate diffusion resistance but may provide insufficient active sites.

Thermal expansion mismatch between metallic TPMS substrates ($15–20 \times 10^{-6}$ K^{-1} for stainless steel) and ceramic catalyst layers ($5–10 \times 10^{-6}$ K^{-1} for alumina) generates severe stresses during thermal cycling. These stresses cause spallation, cracking, and loss of catalytic material—a failure mode observed in multiple studies [141]. The complex curvature of TPMS surfaces exacerbates these stresses compared to flat substrates.

Future directions: Development of electrophoretic deposition methods for uniform coating; exploration of intermediate bond coats to accommodate thermal expansion mismatch; in-situ coating formation via hydrothermal synthesis; computational modeling of coating distribution; and accelerated thermal cycling tests to quantify coating durability.

7.3. Computational and design challenges

The geometric complexity that gives TPMS their advantages also creates severe computational difficulties. Standard CAD environments are optimized for boundary representations rather than implicit mathematical surfaces, necessitating conversion of continuous equations to discrete polygonal meshes. This tessellation generates massive file sizes (often exceeding 1 MB per unit cell) and introduces geometric artifacts that corrupt subsequent simulations.

Mesh generation for computational fluid dynamics of TPMS structures is notoriously problematic. The high curvature and thin walls produce poor-quality elements, leading to convergence failures or inaccurate results. Many studies report meshing difficulties and extensive manual intervention [72,106]. Multi-physics simulations coupling fluid flow, heat transfer, and reaction kinetics across multiple unit cells remain computationally prohibitive for design optimization.

The absence of standardized design guidelines and performance correlations forces case-by-case analysis. Unlike traditional heat exchangers with well-established correlations, TPMS designers must rely on bespoke simulations or experiments for each new geometry. The correlations presented in this review are topology-specific and lack validation across the full parameter space.

Future directions: Development of reduced-order models and surrogate modeling approaches to accelerate simulations; creation of open-access databases of TPMS performance; machine learning-assisted design optimization; standardized meshing protocols; and extension of correlations to industrially relevant operating conditions.

7.4. Operational and stability challenges

The long-term durability of TPMS reformers under realistic operating conditions remains largely unexplored. Most studies report short-term experiments (hours to days) at laboratory scale, whereas industrial applications require thousands of hours of stable operation. Thermal cycling—inevitable in on-board applications with start-stop operation—induces fatigue in both the metallic substrate and the catalytic coating.

Coke formation presents a particular concern for hydrocarbon reforming. The tortuous channels of TPMS structures may create stagnant regions where carbon precursors accumulate, leading to filamentous carbon growth that can block channels and deactivate catalysts. While some studies report coke resistance [141], systematic investigations across fuels and operating conditions are lacking.

Channel blockage by particulates or debris poses another operational risk. The small hydraulic diameters that enable compactness also make TPMS reformers susceptible to fouling. Unlike packed beds that can tolerate some non-uniformity, a single blocked channel in a TPMS structure may render a significant fraction of the reactor volume inaccessible.

Future directions: Long-duration (1000 + hour) testing under cyclic conditions; in-situ diagnostics for coke detection; development of coke-resistant catalyst formulations specifically for TPMS geometries; filtration and pre-treatment strategies for feed streams; and failure mode analysis for industrial deployment.

7.5. Economic and scalability challenges

The economic viability of TPMS reformers remains uncertain. Current fabrication costs for metal additively manufactured components are 10–100 times higher than those of conventional manufacturing per unit volume. While superior performance may justify premium pricing for specialized applications (aerospace, portable power), cost reduction is essential for widespread adoption.

Scalability presents parallel challenges. The maximum build volume of metal additive manufacturing systems (typically 250–400 mm) limits reactor size. Assembling multiple printed modules introduces sealing challenges and increases costs. Throughput remains low: a single reformer may require days of printing time, making mass production impractical with current technology.

The value proposition varies dramatically by application. For on-board hydrogen production in vehicles, the compactness and weight savings of TPMS reformers may offset higher costs. For stationary applications where footprint is less constrained, conventional designs may remain economically superior.

Future directions: Techno-economic modeling comparing TPMS reformers with conventional alternatives; development of high-throughput manufacturing methods (e.g., binder jetting); modular design strategies for scaling; and identification of niche applications where TPMS advantages justify premium costs.

7.6. Synthesis and outlook

The challenges outlined above should not obscure the transformative potential of TPMS-based reformers, but rather frame a realistic research agenda. The path forward requires integrated efforts across multiple disciplines: materials scientists must develop coatable, durable catalyst systems; manufacturing engineers must reduce costs and improve quality; computational modelers must create efficient design tools; and chemical engineers must validate performance under realistic conditions.

Priority research directions emerging from this analysis include: (1) development of robust, scalable coating methods for complex geometries; (2) long-term durability studies under thermal cycling and coking conditions; (3) reduced-order models enabling rapid design optimization; (4) cost reduction through advanced manufacturing and material selection; and (5) demonstration at relevant scale in target applications (on-board reforming, exhaust heat recovery).

The convergence of geometric optimization, advanced manufacturing, and catalytic science positions TPMS structures as a platform technology for process intensification. Realizing this potential requires acknowledging and systematically addressing the substantial challenges that separate laboratory demonstrations from industrial reality.

8. Conclusion

This review has comprehensively analyzed the application of Triply Periodic Minimal Surface (TPMS) structures as a foundational technology for developing compact and highly efficient chemical reformers. The unique geometric properties of TPMS – characterized by zero mean curvature, high surface-to-volume ratios, and continuous, non-intersecting flow channels – render them exceptionally suitable for process intensification. The evidence synthesized from recent studies unequivocally demonstrates that TPMS-based reformers offer significant advantages over conventional tubular and packed-bed designs, including:

1. Enhanced transport phenomena: TPMS structures disrupt thermal and hydrodynamic boundary layers, leading to a substantial augmentation of heat and mass transfer. This results in higher overall heat transfer coefficients and more efficient delivery of thermal energy to active catalytic sites, which is critical for endothermic reactions like steam reforming and ammonia decomposition.

2. Superior catalytic performance: When used as structured catalyst supports, TPMS geometries mitigate internal diffusion limitations common in traditional porous catalyst pellets. The application of thin, uniform wash-coats ensures high catalyst utilization, enabling higher conversion efficiencies at lower temperatures and reducing the required catalyst loading.
3. Compactness and design flexibility: The innate cellular architecture of TPMS, combined with the capabilities of additive manufacturing, allows for the creation of highly compact reformers that can be tailored to fit specific spatial constraints. This is paramount for mobile applications such as on-board hydrogen production and exhaust heat recuperation systems in internal combustion engines and gas turbines.
4. Proven application potential: Experimental and numerical studies across various reforming processes – including ammonia decomposition, methanol steam reforming, and methane reforming – consistently show that TPMS structures (particularly Gyroid and Schwarz D) achieve superior conversion rates and hydrogen yields compared to traditional counterparts.

Despite the promising results, the widespread industrial implementation of TPMS-based reformers requires further research. Key future directions include: optimizing the trade-off between pressure drop and heat transfer for specific TPMS topologies; developing standardized, high-throughput methods for applying uniform and adherent catalytic coatings on complex TPMS surfaces; conducting long-term stability and durability studies under real operating conditions; and exploring the integration of TPMS reformers with advanced systems like membrane reactors and electrified heating.

In conclusion, TPMS-based structures represent a paradigm shift in the design of compact chemical reformers. TPMS-based structures synergize geometric optimization, advanced manufacturing, and catalytic science. They offer a powerful pathway for achieving the high efficiency, compactness, and flexibility required for future decentralized and on-board hydrogen production. Consequently, they will play a pivotal role in the transition to a sustainable hydrogen economy.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Dmitry Pashchenko reports that administrative support was provided by Guangdong Technion-Israel Institute of Technology. Dmitry Pashchenko reports a relationship with Guangdong Technion-Israel Institute of Technology that includes employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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